How China Will Win the AI War

The Convergence Strategy: A Structural Path to AI Dominance

CONDENSED EDITION

A short-form companion to the living book[1]

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[1]This condensed edition abridges the full living book of the same version and date. Every number in it is taken from the full edition, which carries the sources, evidence grades, and derivations. Version v0.20 — 18 August 2026.

About This Edition

The full edition of this book runs to nearly three hundred pages because its argument is a systems argument, and a systems argument has to be shown layer by layer before it can be believed: the model, the compute, the semiconductor, the watt, the data, physical production, trust, governance, finance, and the international order that all of them together imply. Many readers do not need the layers. They need the shape—what is being claimed, why, what would prove it wrong, and what it means for a government, a laboratory, or a company that is neither in Washington nor in Beijing. This condensed edition is for them.

Three rules govern the abridgement. First, nothing here is new: every number, grade, forecast, and recommendation is taken from the full edition of the same version and date, which carries the sources, the evidence grades, the derivations, and the falsification apparatus. Where the condensed text states a figure, the full edition states where it came from and how much to trust it. Second, the industrial precedents that occupy Part I of the full edition—the solar, battery, and electric-vehicle wars, and the campaigns where the same playbook has failed—are here summarized in a single chapter, because for the purposes of the argument what matters is the playbook they establish and its limits, not the campaign histories. Third, the weight of this edition is shifted deliberately toward the overall strategy and its international consequences: what the convergence means for the West, for Japan and Europe, for the non-aligned majority of the world, and for the institution the analysis keeps arriving at. Readers who want the engineering—the memory-architecture arithmetic, the appliance cost model, the sizing framework, the security fault tree—will find pointers into the full edition throughout, marked [full edition, Ch. n].

The title is deliberately provocative, but it is not a prediction of inevitability; it is a structural analysis. “Winning” means what it meant in solar, batteries, and electric vehicles: capturing the overwhelming majority of global production, setting the cost curve, controlling the supply chain, defining the de facto standards, and reducing erstwhile leaders to protected niches. By that definition China has already won three technology wars in which the West began with every advantage. The claim of this book is that artificial intelligence is following the same trajectory, and that the mechanism is visible only when the elements are laid side by side.

A note on the author’s position is owed and is given in full in the preface to the full edition. In brief: I direct the RIKEN Center for Computational Science, whose flagship program runs on an American accelerator architecture and whose deepest partnerships are with American and European institutions. Nothing in my professional position gives me an incentive to argue that the Chinese stack is winning; a great deal of it gives me an incentive to argue the opposite. I did not begin this project expecting to arrive where it has arrived, and I have tried to hold the argument to a standard of proof appropriate to how much I would prefer it to be wrong. Where the evidence does not support the strong claim, the book—and this abridgement of it—says so.

The currency date of this edition is 18 August 2026, the date on the title page. Every factual claim is current as of that date; anything learned after it belongs to the next edition. This is a living document. Corrections are welcome, and the fastest way to change its conclusions is to falsify an entry in the full edition’s evidence ledger.

Contents

Chapter 1

The Thesis

In one paragraph

Between roughly 2005 and 2025 China fought and won three complete technology wars—solar photovoltaics, lithium-ion batteries, and electric vehicles—by executing a single, learnable playbook. Artificial intelligence is the fourth run of that playbook. The mechanism is not to match the West chokepoint for chokepoint but to engineer the chokepoints out of the system: open-weight frontier models released at marginal prices, an open hardware ecosystem built on an unrevocable instruction set and a memory-capacity arbitrage that substitutes abundant commodity DRAM for scarce HBM, and a domestic semiconductor base that removes the last physical dependencies. The connective tissue is openness deployed as a weapon against the layers where Western incumbents collect their rents. The argument is decomposed into four propositions that are graded and falsified separately, and the book states in advance what would prove it wrong.

1.1 Three wars, one playbook

In solar photovoltaics China went from assembler of imported cells to producing more than 90% of every stage of the global supply chain, while module prices fell by more than 99% from their 1976 level. In lithium-ion batteries—a technology commercialized by Sony in 1991 and led for two decades by Japan and Korea—Chinese firms now hold roughly 69–75% of global electric-vehicle battery installations and over 80% of global cell manufacturing capacity, with pack prices down 93% in real terms since 2010. In electric vehicles China became simultaneously the world's largest car market, the world's largest car exporter, and home to the world's largest EV maker, while foreign incumbents' share of the Chinese market collapsed from 62% to 35% in five years.

These were not three lucky outcomes. They were three executions of one playbook, whose elements Chapter 2 distills: patient state designation of a strategic industry; demand creation at home; subsidized scale-up of manufacturing far ahead of demand; brutal domestic competition that functions as an evolutionary selection engine; capture of the entire upstream supply chain; a cost-down learning-curve war that bankrupts foreign incumbents; and finally, once dominance is achieved, the conversion of that dominance into geopolitical leverage—export controls on the very technologies China was once accused of copying.

1.2 Why AI is the fourth war

The prevailing Western assumption is that AI is different: that frontier AI depends on scarce, controllable inputs—leading-edge logic, high-bandwidth memory, proprietary interconnects, closed model weights—and that export controls on these chokepoints can durably preserve a lead. The book argues that the assumption is failing on all fronts simultaneously, and failing in a characteristic way: not because China matches the West chokepoint for chokepoint, but because it is engineering the chokepoints out of the system.

The model. Chinese laboratories have converted the model layer into a commons. As of mid-2026, Chinese open-weight models occupy the top open positions on independent intelligence indices, account for roughly 41% of all Hugging Face downloads and a majority of tokens routed through neutral API aggregators, and are priced at a small fraction of Western closed models—while the measured capability gap to the best closed US models has compressed to low single digits. Open weights at marginal cost do to the model layer what Chinese modules did to solar: they set a world price that rent-seeking business models cannot survive.

The compute. The state has designated the open RISC-V instruction set as national infrastructure, is standardizing matrix extensions and chip-to-chip interconnects across vendors, and has open-sourced the software stacks that would otherwise fragment the domestic ecosystem. In parallel, a memory-capacity arbitrage—substituting terabytes of cheap commodity DRAM for scarce HBM—attacks the assumption that frontier training requires centralized, HBM-bound superclusters at all.

The semiconductor. Beneath both, the fabs. SMIC has reached 5 nm-class logic without EUV; CXMT has become the world's fourth DRAM producer and the swing supplier in a global memory shortage; the domestic equipment industry has placed three firms in the global top twenty. The remaining gaps—EUV lithography, leading-edge HBM—are real, but the strategy does not require closing them on the West's terms; it requires making them irrelevant to the chosen architecture.

1.3 Openness as a weapon

The connective tissue among the elements is a deliberate inversion of the Western technology-rent model. Where the United States concentrated AI value in proprietary layers—CUDA, NVLink, closed frontier weights, x86/Arm licensing—China's strategy, formalized in the July 2025 Global AI Governance Action Plan and the "AI+" State Council initiative, is to make every one of those layers open, standardized, and free. This is not altruism; it is the same move as pricing solar modules at $0.09 per watt. An open layer generates no rent for the incumbent, but it generates *adoption*, and adoption generates the scale, the ecosystem, and the standards-setting power that constitute victory in the precedent industries. The West's chokepoint strategy and China's commoditization strategy are not symmetric opponents: one defends margins, the other destroys them.

1.4 Four propositions, stated separately

Rigor requires decomposing the thesis into four propositions that are related but logically independent—none proves the next, and each has its own evidence and its own failure modes.

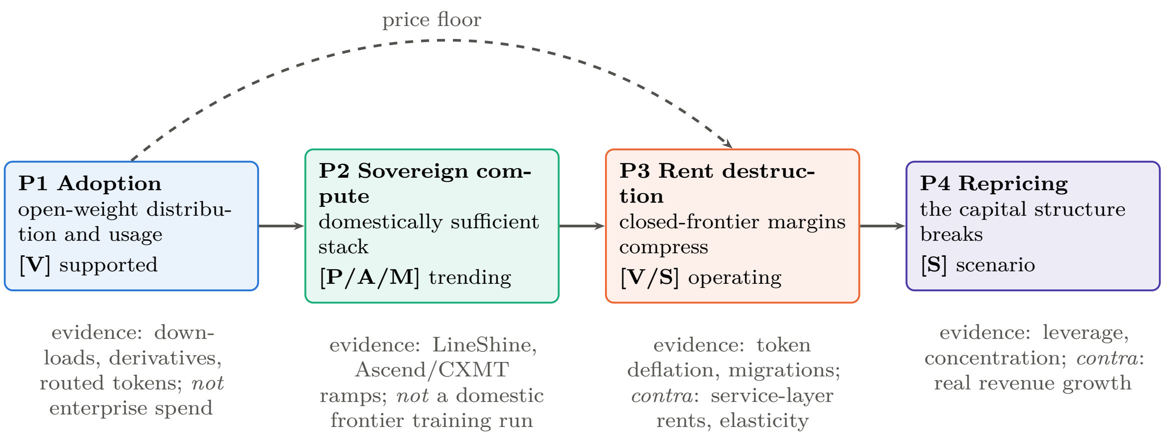
P1 — Adoption. China is winning open-weight model adoption. *Confirmed by* download, derivative, and routed-token majorities persisting and extending into enterprise spend. *Falsified by* a durable reversal of the router and Hugging Face shares, or enterprise open-weight spend stagnating below about 15% while usage plateaus. Current grade: strongly supported [V].

P2 — Sovereign compute. China is constructing a domestically sufficient training-and-serving stack. *Confirmed by* a frontier-class model trained end-to-end on domestic silicon, packaging, and memory. *Falsified by* persistent dependence on stockpiled or smuggled foreign accelerators and HBM past about 2028. Current grade: rapid progress, not yet achieved [P/A/M].

P3 — Rent destruction. Open-weight commoditization will compress the closed-frontier rents that finance Western laboratories. *Confirmed by* closed-model revenue growth decelerating while capability parity holds. *Falsified by* closed labs sustaining pricing power through service, distribution, and trust layers even at model parity. Current grade: mechanism operating, magnitude contested [V/S].

P4 — Financial repricing. The compression of P3, if it occurs, punctures the US AI capital structure. *Confirmed by* the indicator table of Chapter 5. *Falsified by* demand elasticity absorbing the price decline: usage growing faster than prices fall rescues the buildout even under P1–P3. Current grade: scenario judgment [S].

Figure 1.1 shows the propositions with their couplings and their contrary evidence. P1 can hold while P2 fails (China dominating distribution on stockpiled NVIDIA silicon); P3 can hold while P4 fails (margins compress while volume growth fills the data centers); and a US valuation correction could arrive from macro causes with none of P1–P3 mattering. The convergence argument is that P1 is established, P2 is trending, and the *coupling* of P1×P2—open weights meeting sovereign, capacity-first hardware—is what forces P3, with P4 as its priced-in consequence.



None of these arrows is entailment. P1 can hold while P2 fails (dominance on stockpiled foreign silicon); P3 can hold while P4 fails (margins compress while volume growth fills the data centres, the elasticity case); and P4 could arrive from macro causes with none of P1–P3 mattering. The convergence argument of this book is that the *coupling* P1×P2—open weights meeting sovereign, capacity-first hardware—is what forces P3, with P4 as its priced-in consequence.

Figure 1.1: The argument, disaggregated. Each proposition carries its own evidence grade, its own confirming metrics, and its own contrary evidence; the book's thesis is a claim about the coupling between them rather than a single chain of entailment. Grades as of the version stamp; the full edition's evidence ledger carries the full ledger and its dashboard the quarterly indicators that move them.

1.5 Evidence grades

Load-bearing claims are graded throughout: **[V]** independently verified; **[P]** primary-source claim (reliable as to what was claimed, not what was achieved); **[A]** analyst estimate; **[M]** author-modeled result; **[R]** roadmap; **[S]** scenario judgment. The full edition collects every headline claim in a graded ledger and a machine-readable claim register [full edition, App. C, I]; Appendix A of this edition reproduces the forecasts and the quarterly dashboard.

1.6 The theory, in brief

The full edition devotes a chapter to the machinery behind the analogy [full edition, Ch. 2]; four ideas from it are needed to read what follows.

A cost stack, not a price. The cost of one completed useful task decomposes into amortized model creation, inference, integration and organizational adoption, complementary capital, and assurance. Only inference is falling at ten times per year. Integration and adoption are not falling—which is why open-weight *usage* share rose while open-weight *spend* share fell—and assurance is a cost the commons has mostly not paid. Commoditizing inference therefore does not commoditize delivered work; it shifts the binding constraint to the terms that are not falling. The rent-destruction claim is bounded to the model layer and remains a hypothesis at the service layer.

The elasticity knife-edge. Whether spending on AI rises or falls when its price falls is the whole financial argument stated without mathematics. With own-price elasticity ε, revenue moves as $\Delta \log R = (1 - \varepsilon)\, \Delta \log P$: above one, price cuts pay for themselves; below one, falling prices shrink revenue; $\varepsilon = 1$ is the boundary between the bull and bear cases for the entire Western AI complex. The honest position is that ε is a portfolio, not a number—plausibly above one for coding agents, synthetic data, and scientific inference, near or below one for consumer chat—and that the leveraged Western structure needs it above one at the premium tier specifically. Graceful degradation is a real Chinese advantage; it is not a theorem of victory.

Openness relocates rent; it does not abolish it. A seller rationally prices at or below marginal cost whenever the extra volume raises revenue on a complement by more than the margin given up. Each side opens the layer where it is behind and expects rent in the layer it is best at: China opens weights, ISA, interconnect, and framework and expects to recapture on hardware volume, domestic cloud, standards, and dependence; the United States controls all four and expects premium capability, distribution, integration, trusted deployment, and agent platforms. Two flywheels result, and each can steal the other's critical input—China steals volume, America steals value. Which theft dominates is, on the book's reading, the single most important unresolved empirical question in the field.

Four layers of winning. AI is a general-purpose technology, which means a country can dominate its production and capture little of its value, because most of the value appears downstream in diffusion and productivity. Production leadership is the only layer on which the United States is unambiguously ahead. The platform layer is going to whoever opens fastest; diffusion is where most realized value sits; productivity is the only layer that finally matters and the least observable in real time. A country can lead one layer while losing another.

The thesis, stated in its conditional form: the United States retains the absolute model frontier, the leading semiconductor and cloud ecosystem, and vastly greater private capital. China increasingly leads a different contest—the commoditization of frontier-adjacent intelligence and the construction of a sovereign, though incomplete, compute stack. The decisive question is whether Chinese cost reduction, adoption, standards formation, and hardware sufficiency compound faster than US capability, distribution, and monetization—and whether China closes its remaining gaps in memory, packaging, tools, software maturity, and trust before the open-model economy compresses the rents that finance the Western frontier. This book argues that convergence is the way to bet, and Chapter 8 states what would change the bet.

Chapter 2

The Precedents and the Playbook

In one paragraph

Solar showed the pure form: Western invention and Western demand, Chinese policy-financed scale, capture of the learning curve, failed tariffs, a home market that made foreign trade policy irrelevant, more than 90% share at every stage, and then state cartelization. Batteries added vertical depth from the mine to the pack, won by regulatory market reservation rather than tariffs and by industrializing the "inferior" chemistry until the West licensed it back. Electric vehicles proved the playbook works on a whole system, not just a component, and turned the car into a Chinese-led embedded-AI platform. From the three campaigns the book distills nine tactical moves and six strategic steps, and from the campaigns that have *not* worked—aircraft, lithography, precision tools, operating systems—it distills where the playbook grinds: tacit knowledge, slow clock speed, foreign certification, installed-base lock, and trust. Scored against that model, AI's model and compute layers sit in the kill zone; its deployment layer is genuinely contested.

2.1 Solar: the template

The silicon solar cell was invented at Bell Labs in 1954, and the industry was built on American, German, Japanese, and Australian science; Sharp was the largest cell producer in 2006. Demand was created by Western policy, above all Germany's 2004 feed-in tariff. The pipeline into China ran through Australia: Shi Zhengrong, a student of Martin Green at UNSW, founded Suntech in Wuxi in 2001 with municipal seed capital. Chinese scale-up rode Western tariff demand and policy credit—the China Development Bank extended roughly $47 billion of credit lines to renewables manufacturers in 2010, the year solar was designated a Strategic Emerging Industry. The polysilicon price collapsed from about $475 per kilogram in 2008 to under $16 by the end of 2012, and Chinese capacity annihilated Western and Japanese incumbents built for high-price niches: Solyndra, Q-Cells, SolarWorld. Crucially, the shakeout also killed Chinese pioneers—Suntech defaulted in 2013—because the state protects the industry, not the champion.

Western trade defenses failed. US anti-dumping and countervailing duties in 2012, EU duties in 2013 settled within weeks for a minimum-price undertaking, and the 2018 US safeguard tariff all taxed deployment without resurrecting manufacturing; Chinese producers moved assembly to Southeast Asia and moved up the value chain, and by the time the tariff walls were complete there was no meaningful Western industry left inside them to protect. From 2011 China built a domestic market that dwarfed exports and insulated its manufacturers from foreign trade policy: 315 GW_{AC} installed in 2025 alone, roughly 60% of world additions, and a cumulative 1.2 TW.

The end state is total supply-chain capture: 2023 Chinese shares of 92% in polysilicon, 98% in wafers, 92% in cells, and 85% in modules, at costs 10–35% below India, the United States, and Europe, with global module capacity of about 1.8 TW a year against demand of 0.6–0.7 TW.

Tariff-protected markets now import the layer below the tariff line. The cost curve is the weapon: module prices fell from $106 per watt in 1976 to about $0.09 today at a learning rate near 20% per doubling, and from roughly 2010 onward essentially all doublings were Chinese—whoever manufactures the doublings owns the learning (Figure 2.1). Winning is expensive: prices below cash cost, forecast 2025 losses of RMB 29–33 billion at the five majors, and then Xi's July 2025 "anti-involution" campaign, a capacity-buyout fund, and a polysilicon "storage and integration platform"—an OPEC for solar. Having captured the industry, the state now moves to cartelize it and harvest stable rents. The Western rebuild did not happen: the Inflation Reduction Act built module assembly but not cells, its successor legislation cut deployment forecasts, and First Solar survives as a tariff-protected niche shipping about one fortieth of Chinese volume—the terminal scoreboard of the first war.

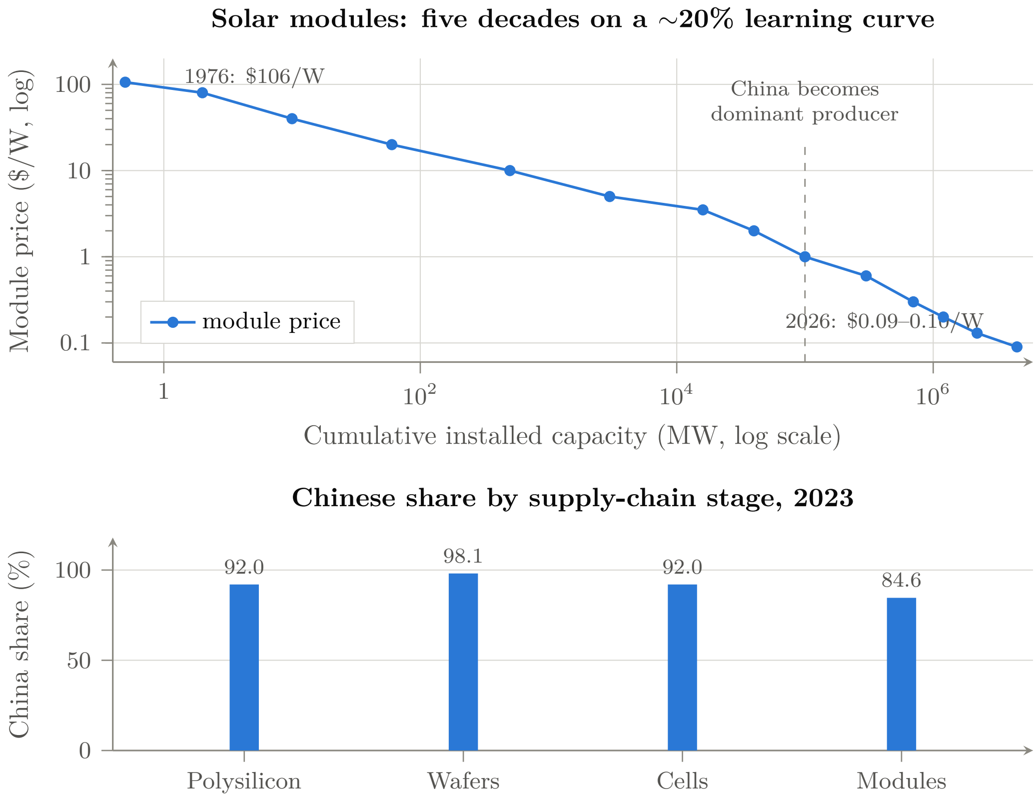


Figure 2.1: The template war. *Above*: module price against cumulative installed capacity, both log scales—a Wright's-law decline of roughly 20% per doubling, from $106/W in 1976 to $0.09–0.10/W in 2026. Whoever manufactures the doublings owns the learning; from about 2010 essentially all doublings were Chinese. *Below*: 2023 production shares by stage. Values [V]; the price series mixes contemporaneous reporting with index reconstruction and is directional at any single point.

Four lessons carry forward. Trade barriers imposed after cost-curve capture protect nothing but domestic prices. Champions are disposable; the industrial commons is not. A subsidized home market large enough to absorb half of world demand makes foreign trade policy irrelevant. And the endgame of commoditization is state-managed consolidation—dominance first, margins later. Keep all four in mind when reading about open-weight models priced near zero.

2.2 Batteries: vertical depth

Sony commercialized the lithium-ion cell in 1991; Japan and then Korea led for two decades. China entered through low-margin consumer electronics—BYD was the world's largest nickel-cadmium maker by 2002; ATL made phone cells and spun out CATL in 2011—and pivoted to the strategic segment when the state created demand. The decisive instrument was not a tariff but a certification: the 2015 "white list" of 57 domestic battery makers whose cells qualified for new-energy-vehicle subsidies, which excluded LG and Samsung just after they had built Chinese plants, and which was scrapped in June 2019 precisely when the champions no longer needed it. Regulation as industrial policy, adjusted dynamically, with foreign firms as designated losers.

The end state in 2025: CATL 39% and BYD 16% of a 1.19 TWh market, Chinese makers roughly 69% of installations and over 80% of the world's 4 TWh of cell capacity against 6–7% each for the EU and the United States; upstream, 85% of anodes, 82% of electrolytes, 74% of separators, 70% of cathodes, over 90% of graphite processing, and two thirds of lithium and cobalt refining. Pack prices fell from $1,474 per kWh in 2010 to $108 in 2025, with a 44–56% penalty for packs made in North America or Europe. A competitor who builds a gigafactory outside China still builds it, in effect, inside the Chinese supply chain.

The most instructive episode is lithium iron phosphate. Invented in the West, dismissed as an energy-density dead end, LFP was rescued by Chinese pack engineering and passed 55% of the global EV market in 2025; Ford's Michigan plant runs under a CATL license and CATL and Stellantis are building an LFP plant in Spain. The "inferior" technology branch, industrialized ruthlessly at scale, beat the premium branch on system economics—a preview of the LPDDR-versus-HBM argument in Chapter 3. The arc completed in 2025 when the infant industry became the gatekeeper: MOFCOM export licensing on LFP cathode process technology, then on high-energy cells, anodes, and equipment—the mirror image of US chip controls. Northvolt, with $15 billion raised, went bankrupt having produced about one five-thousandth of CATL's output; grid storage was decided before most Western strategy documents mentioned it. Three more lessons: regulatory market reservation is more effective than tariffs and invisible to trade law; vertical integration to the mine makes a foreign competitor's factory a hostage, not a rival; and the dismissed-technology branch is where cost-curve wars are won—watch what China industrializes, not what the West benchmarks.

2.3 Electric vehicles: the system-level win

The car is the most complex mass-manufactured product in existence, wrapped in brands, dealers, safety regulation, and a century of incumbency; if the playbook worked here, no consumer-industrial market is safe from it. It worked here. The strategic decision predates Tesla: Wan Gang persuaded Beijing around 2000 that China could never catch Toyota and Volkswagen in combustion but could leapfrog by changing the energy carrier. Early demand instruments mostly failed; the state persisted anyway. Cumulative support of at least $231 billion through 2023 was layered as purchase subsidies, tax exemptions, free plates, the 2017 dual-credit mandate that forced every maker including foreign joint ventures to build NEVs, and 19 million charging points, 4.6 million of them public—about eighteen times the US public network. The result was the fastest substitution event in automotive history: NEV penetration from about 6% in 2020 to 54% in 2025, China the largest car exporter since 2023, and monthly exports past a million vehicles by mid-2026.

BYD is the pattern's exemplar: a battery maker that bought a carmaker in 2003, spent fifteen unremarkable years, then ran the flywheel from 416,000 vehicles in 2020 to 4.55 million in 2025, roughly 80% of Tier-1 content in-house, a structural cost advantage over Tesla of about $4,700 per vehicle of which direct subsidy is under $300, its own ro-ro fleet, and plants inside every tariff wall. Around it an ecosystem of siege, and behind it a graveyard: some 400 Chinese

EV ventures died between 2018 and 2025. The state subsidizes the species, not the specimen. Foreign brands' share of the Chinese market fell from 62% to 35%, and the technology now flows backward: Volkswagen licenses XPeng's autonomy stack, Audi builds on a SAIC platform, Stellantis owns a fifth of Leapmotor, Toyota's Chinese BEVs carry BYD powertrains. A century of licensing ran the other way; it took one product generation to reverse it. The Western trade response—100% US tariffs, EU duties being negotiated toward a minimum price while BYD builds in Hungary and Turkey—repeats the solar script.

The bridge to Part II is that the car became a software and AI platform in Chinese hands: Chinese suppliers ship about 95% of automotive lidar, city-level autonomous navigation runs at scale from 2024, driver assistance ships at zero cost on an $8,000 Seagull, and McKinsey finds Chinese vehicles "frequently more advanced" with 20–24-month development cycles against 40–50. The first mass-market, safety-critical, embedded-AI product category is already Chinese-led, and it is the demand anchor for the domestic AI chips, sensors, and models of the next chapter. Three final lessons: pick the transition where incumbency resets to zero and force it by regulation at home; a system-level product won end to end pulls its whole component stack to dominance with it; and speed is a weapon—a twofold development-cycle advantage compounds every generation.

2.4 The playbook distilled

Table 2.1 inventories nine tactical moves that recur, in different orders, across the campaigns; the AI column is developed in Chapters 3–4. Six strategic steps are the sequence those moves serve. *Target selection*: an industry at a technology discontinuity where the incumbent's accumulated advantage—combustion drivetrains, HBM-centric clusters—is about to be stranded, and where manufacturing learning-by-doing rather than protected IP is the true accumulating asset. *Demand before supply*: the home market as guaranteed demand, with foreign firms admitted just long enough to transfer capability. *Overcapacity as strategy*: capacity funded at multiples of world demand, which Western economics reads as waste and which strategically purchases the entire learning curve, a permanent price umbrella too low for any entrant, and surge capacity; the losses land on Chinese balance sheets and dead firms while the cost curve remains. *Selection, not planning*: Beijing floods the sector and lets domestic combat select; its loyalty is to the industrial commons—the engineer pool, the supplier mesh, the tooling ecology. *Commoditize the adversary's rent layer*: wherever the Western business model collects rent—module, pack, badge, API token, ISA license—price it at marginal cost, because rents fund the West's R&D flywheel and destroying the rent destroys the flywheel; openness is this move in its purest form, since the rent goes to zero by construction and adoption accrues to the commoditizer. *Endgame*: consolidate, stabilize prices upward, and convert dominance into coercive leverage through export controls on the now-Chinese-controlled layer.

Against this the West has fielded two counter-patterns, both failing. Tariff walls arrive after cost-curve capture and merely tax domestic consumers while the walled market shrinks in relevance. Chokepoint denial—semiconductor export controls since 2019—is the more serious strategy, and its record is a pattern the book calls *chokepoint decay*: each denied input (EUV, HBM, H100s, EDA) triggers a state-scale substitution program plus an architectural detour around the input, while the denial itself hands the domestic substitute a guaranteed market—the white-list mechanism, imposed this time by Washington on Beijing's behalf. Export controls are, functionally, a US-funded market-reservation policy for Huawei, SMIC, CXMT, and Cambricon. Whether the decay outruns the West's lead is the empirical question of the next chapter.

2.5 Where the analogy breaks

A playbook argued from three victories owes the reader its defeats. The same state, spending an estimated $248 billion a year on industrial policy, has run two-decade campaigns where

Table 2.1: The nine tactical moves, across four campaigns.

Move	**Solar**	**Batteries**	**EVs**	**AI**
M1 State designation	SEI 2010, 12th FYP	White list 2015	863 program, SEI 2010	“AI+” plan, 15th FYP, RISC-V guidance
M2 Home demand	National FiT, Top Runner	NEV subsidy linkage	\$231B support, dual credit	State DCs mandated on domestic chips, subsidized power
M3 Subsidized scale-up	CDB \$47B credit	Provincial capital	Provincial EV funds	Big Fund III (\$47.5B), SMIC/CXMT capex
M4 Darwinian competition	2018 “531” cull	100+ makers to consolidation	∼400 dead EV firms	LLM price war since 2024; chip-vendor shakeout beginning
M5 Supply-chain capture	poly → module	mine → cell	battery → chip → magnet → ship	model → framework → chip → interconnect → fab → DRAM
M6 Commoditize the rent layer	module \$/W	pack \$/kWh	vehicle BOM	open weights, \$/Mtoken, open ISA and stack
M7 Absorb the “inferior” branch	mono-PERC, TOPCon	LFP	small BEVs	sparse MoE; LPDDR capacity vs HBM bandwidth
M8 Standards capture	efficiency floors	GB safety standards	charging standards	RISC-V profiles, matrix ISA, UnifiedBus
M9 Leverage after victory	OPEC-for-poly	tech export controls 2025	rare-earth controls 2025	(pending)

dominance has not arrived. Commercial aircraft: COMAC’s C919, nineteen years after launch, delivered about thirteen aircraft in 2025 against a plan of 75, carries more than 40% Western core content including an engine that is nearly a third of its cost, and saw that engine’s export licenses suspended and restored in 2025—chokepoint leverage of the kind that failed in solar, working. Lithography and the deep tool chain: under 5% localized, a domestic immersion tool resembling ASML’s 2008 generation, photoresist and vacuum valves under 10% domestic—twenty years and three Big Funds have not cracked it. Precision machine tools and reducers: five-axis CNC about 8% domestic against an 80% target, harmonic reducers about 85% Japanese as of 2023, though the siege progresses. Operating systems: two decades of state Linux never dented Windows; HarmonyOS reached 17% of the mainland smartphone market only after US sanctions burned the Android bridge—ecosystems yield eventually, but on decade timescales, under existential forcing, and so far only inside the protected home market. Biotech is the surprise late bloomer, with \$138 billion of out-licensing in 2025 and a residual gap that is entirely trust and regulation: the FDA’s 14–1 rejection of China-only trial data is the certification barrier in its purest form. Biotech in 2025 looks like batteries in 2015, which is exactly why the trust chapter matters for AI.

The variables separating success from failure are consistent: the playbook wins where products are modular, iteration is fast, learning is manufacturable, the home market can certify, and trust is embodied in the artifact; it grinds where knowledge is tacit, certification is foreign, and value lives in an installed ecosystem. Scored against that model, AI models are the most modular, fastest-clock-speed artifact in industrial history—zero-cost weights, months-long frontier turnover, benchmark-verifiable, codified and published recipes; sparse-MoE training is closer to batteries than to jet engines. Compute is intermediate: chiplet products iterate quickly, while the tool chain inherits lithography’s resistance, which is why the strategy routes around it. But three properties of AI sit in the resisting column, and they are the strongest structural arguments against this book’s title. The frontier moves: a fast follower is never finished, and recursive AI-for-AI research could reopen the gap faster than any factory closes it. Ecosystems lock: the enterprise deployment layer is Office, not modules, and it is the closed labs’ real moat. And trust and certification bind: models in government, finance, and defense are like avionics or drugs, and the sintilimab precedent—capability accepted, certification denied—is directly available to

Western regulators as an AI playbook. Two asymmetries follow. Zero reproduction cost lets Chinese weights conquer adoption instantly, and would let a single superior Western open release reconquer it just as fast; there is no factory-shaped moat around P1. And talent is the one input where the two systems still share a supply chain: researchers of Chinese undergraduate origin were 47% of the top tier in the 2022 cohort, up from 29% in 2019, and the share of that tier working in the United States fell from 59% to 42% while China's rose from 11% to 28%. The honest conclusion is that the model and compute elements sit mostly in the kill zone, the tool chain in the resistance zone and bypassed, and the deployment layer—trust, certification, ecosystems—is genuinely contested ground where the precedents predict a long grind, not a collapse. That is where Chapter 4 and the strategy chapters place their weight.

Chapter 3

The Elements of AI

In one paragraph

Read separately, each element has a Western rebuttal: the models trail the closed frontier by a few points, the accelerators trail NVIDIA per watt, the fabs trail TSMC by two nodes. Read together, the rebuttals miss the structure. The model layer has been converted into a commons by a self-enforcing open-weight ratchet; the harness—the runtime that agents are programmed against—is being given away one layer up; a GPU-free Chinese machine leads both TOP500 benchmarks and shows the CPU/accelerator boundary to be an allocation choice, while a memory-capacity arbitrage and a five-thousand-dollar appliance attack the premise of the centralized supercluster; the chokepoints on Chinese silicon are decaying faster than they are replaced, with domestic HBM the one binding constraint; the watt is China's deepest structural asymmetry, aimed by policy at domestic chips; deployment-generated data is where rent migrates when the model becomes free; and embodied AI is where the loop closes back onto the factory floor. None of it is finished, every claim is graded, and the single integration test that has not been passed—a frontier model trained end to end on domestic silicon—is the "Mate 60 moment" the forecasts price at 65% within two to four years.

3.1 The model: open weights as the commodity layer

The Western frontier is closed: the best US models are accessible only as metered APIs, their weights corporate secrets, their pricing a rent on capability. The Chinese frontier is open: downloadable weights, mostly under MIT or Apache licences, at API prices near marginal inference cost. The inversion began with DeepSeek. Its V2 release in May 2024, with multi-head latent attention and fine-grained mixture-of-experts at roughly RMB 1–2 per million tokens, triggered a domestic price war in which ByteDance, Zhipu, Alibaba, and Baidu cut prices by up to 97% within two weeks. V3 in December 2024 reached GPT-4-class capability at a published marginal training cost of $5.6 million—a final-run figure [P], not a program cost; estimates of DeepSeek's buildout run up to $1.6 billion [A]. R1 in January 2025, MIT-licensed, matched o1-class reasoning, topped the US App Store, and erased $589 billion of NVIDIA's market value in a day. The strategic content was mispriced by markets and correctly priced by Beijing: frontier capability can be commoditized; export-control-constrained hardware forced algorithmic efficiency—latent attention, sparsity, FP8, GRPO, multi-token prediction—that compounds on any hardware; and open weights convert capability into global adoption instantly. The defensible causal claim is constraint-consistency, not unique causation: nearly all of these techniques would be commercially attractive without sanctions, and the genealogy is graded accordingly [full edition, Ch. 8].

The state did not anoint DeepSeek; it flooded the sector. By mid-2026 the release lag from the US closed frontier to a Chinese open equivalent had compressed from six to eight months to a matter of weeks, and the breadth of the field is itself the datum: Alibaba's Qwen family (a billion cumulative downloads, more than half of global open-model downloads, over 200,000 derivatives), Moonshot's Kimi K3 (2.8 trillion parameters, the largest open-weight model ever released, in the global top five and the only open model in that tier), DeepSeek V4, Zhipu's GLM-5 trained on Ascend, MiniMax, Xiaomi's 1.02-trillion-parameter MIT model, and Xiaohongshu's dots3-note—frontier-scale development is now an expected capability of any large Chinese platform. Token prices run from one seventh to one hundred-and-fiftieth of US equivalents. Chinese models account for about 41% of Hugging Face downloads and roughly 61% of tokens routed through the largest neutral aggregator, up from about 2% in early 2025; roughly 80% of open-stack startups use a Chinese model. The Stanford AI Index put the top US–China gap at 2.7% in March 2026, down from 17–32% in 2023, and the closed-over-open gap at 3.3%—a 23:1 private-spending ratio buying a 2.7% measured gap.

Two developments in the summer of 2026 changed the character of the commons. The first is the *open-weight ratchet*. Alibaba's Max tier had always been proprietary; seven days after Moonshot opened Kimi K3, Alibaba committed to opening Qwen3.8-Max, and shipped the weights on 13 August. Once one Chinese laboratory opens at frontier scale, the others cannot stay closed—adoption is the currency and closing your best model, in that ecosystem, is unilateral disarmament. This upgrades openness from a state policy that could reverse to a competitive equilibrium from which no participant profits by defecting. But the ratchet holds on weights and fails on terms: the Qwen3.8-Max licence requires a commercial agreement above US$50 million of revenue, and Kimi K3's carries the same structure at a US$20 million threshold. Opening the weights does not abolish rent; it relocates it to large-scale commercial service, where the releasing lab retains a claim—a commons with a toll gate at commercial scale. The second is Meta's return to open weights with Muse Glimmer on 10 August, Apache-licensed and competitive if not commanding, with an open flagship promised. Whether it re-anchors a Western pole for the commons or merely confirms that the ratchet now operates on both sides of the Pacific is registered as a forecast; either way, nobody now argues the commons does not matter.

Three qualifications keep the model chapter honest. Capability, distribution, and revenue are three different scoreboards, and the third lags: enterprise open-weight *spend* share fell from 19% to 11% in 2025 even as usage rose, which is the strongest evidence against P3 and is treated as such. Alibaba's claim that Qwen3.8-Max ran "500-plus turns of chip design optimization" toward an autonomous silicon design flow—if true, an attack on the one input export control cannot embargo at a border—is the vendor's own and unverified. And US closed models are rumoured to be roughly twice the parameter count of the Chinese flagships: this is an efficiency victory, not a scale victory. What quantization adds is operation: dynamic quantization and native four-bit release formats put frontier inference on \$2,000–\$40,000 hardware, the capacity-rich design point discovered bottom-up by hobbyists that Chinese hardware policy is discovering top-down. Open weights alone democratize access; quantization democratizes operation. And the right denominator is cost per solved task, not price per token, because reasoning outputs are inflating about fivefold a year and no vendor publishes the number that matters. In playbook terms the model layer is already in the late-war stage: rent destroyed, adoption captured, standards increasingly set in Chinese repositories. What open weights cannot guarantee is sovereign hardware to train and run them.

3.2 The harness: the layer the contest moved to

Every preceding paragraph treated the model as the object of competition; the object is moving. Models are swappable—an API endpoint exchanged by configuration, a product category exists to make it trivial. A *harness*—the runtime that holds the agent loop, tool schemas, permission system, memory and context management, sandbox, session store, and subagent orchestration—is what an organization writes code against, and its abstractions become the integration, its execution semantics become the test assumptions, its permission model becomes what an auditor reads. The layer where switching is expensive is not the layer everyone measures, and a strategy that wins the harness can lose every benchmark and still take the ecosystem. The sole controlled study found that adopting a better harness with *any* model saved 33–61% of cost per task, where switching from the cheapest to the most expensive model saved 36%; and once a self-hosted open model runs at about a dollar per million output tokens against $25–30 for the closed frontier, the harness is closer to the only lever left on production economics.

On 13 August 2026 DeepSeek released an MIT-licensed, plugin-native, model-agnostic harness that passed 100,000 GitHub stars in about 48 hours—built on the kernel of a four-year-old Chinese chatbot framework whose creator DeepSeek recruited. That is an ecosystem-acquisition strategy, and it is this book's thesis executed one layer up. It shipped the same day DeepSeek raised its API prices: commoditize the complement, monetize the scarce layer, and ensure the commoditization damages competitors' pricing power at least as much as your own—the rent-relocation thesis performed rather than argued. The next day Xiaohongshu shipped an Apache-licensed omni-modal agent model trained for ten-hour agent runs; model and harness are being co-designed the same week by two Chinese labs. Meanwhile the Western agent-standards bodies—the foundation that now hosts MCP, the A2A consortium, the Agent Plugins group—have no Chinese named member, and Beijing's cyberspace administration has published a rival "interoperability" initiative naming none of them. The incumbents convene a consortium; the challenger declines to join, ships a permissively licensed implementation, and bids for the installed base directly. The book grades the strong lock-in claim as a hypothesis [S]—the interface is commoditizing, the architectures have converged, and nobody has measured switching cost—but lock-in relocates rather than disappears: to distribution, to the evaluation substrate (whoever owns the harness owns the conditions of measurement), and to certification. If American models remain the best and the world's agents are orchestrated by runtimes shipped from Hangzhou and Beijing under MIT licences, superiority at the model layer buys less than its owners expect [full edition, Ch. 9].

3.3 The compute: RISC-V, accelerators, and the memory arbitrage

The Western AI machine is a proprietary stack—x86 or Arm hosts, NVIDIA accelerators, CUDA, NVLink, HBM—each layer a licensed chokepoint. China's counter-design is an open machine: an open instruction set, multi-vendor domestic accelerators unified by open interconnect and open software, and a memory system that trades scarce bandwidth for abundant capacity. The claim is deliberately narrow: not that the GPU is obsolete, but that the CPU corner of a design continuum can reach parity on the workloads that matter while carrying an ecosystem advantage the accelerator corner does not have.

RISC-V. In March 2025 eight government bodies drafted national RISC-V policy guidance, and the 15th Five-Year Plan mandates "extraordinary measures" for self-reliance with the open ISA as its core mechanism. Because RISC-V is an open standard, its deployment insulates Chinese silicon design from sanctions at the ISA layer permanently: there is no license to revoke. Twelve of the twenty-four premier members of RISC-V International are Chinese; the matrix-extension task groups now standardizing the operations AI silicon executes are, in the book's phrase, the single most consequential unglamorous venue in the field, and their ratification is only the first rung of a maturity ladder from specification to compiler to kernel library to framework to fleet validation—so the RISC-V accelerated-CPU path is real but dated. Washington's attempts to restrict RISC-V participation produced nothing; the Commerce Department conceded that restriction would harm US firms without slowing China. And the platform owner of the AI era has ported its crown-jewel software to the open ISA at a summit in Shanghai: when NVIDIA brings CUDA to RISC-V, Google and Meta build accelerator control planes on it, and the standards war is not being lost but conceded.

The existence proof. On 24 June 2026 LineShine, at the national supercomputing center in Shenzhen, took first place on both the HPL and HPCG benchmarks with 2.198 exaflops of FP64—the first two-exaflop machine, the first Chinese No. 1 since 2017—and no GPUs. Its LX2 processor is an Armv9 part with 304 cores, scalable vector and matrix extensions in every core, and 32 GB of on-package HBM: Fugaku's lineage carried to its conclusion, the accelerator's defining features absorbed into the CPU. Export controls denied China the discrete GPU; the response was to demonstrate, at No. 1 in the world, that the functional boundary between CPU and accelerator is an implementation choice—and then to choose an implementation with no controllable product category in it. The book is exact about the proof boundary: HPL and HPCG are settled [V]; AI serving parity per terabyte-per-second of bandwidth and a training penalty of 4.5–8.5× that is implementation-fixable are the author's modeled results with Dongarra and Hoefler [M]; and a seven-row proof suite states, benchmark by benchmark, what would settle or falsify the AI claim—one of which is a domestic Chinese submission to MLPerf Training, whose June 2026 round drew 95 systems on thirteen accelerators, none of them Chinese. Nonparticipation is not proof of weakness, but it is a measurable international-validation gap.

The fleet. US bans and Beijing's own demand squeeze took NVIDIA's China data-center revenue from a 95% share to approximately zero; state-funded data centers must use domestic chips. Ascend, Cambricon, Kunlun, and T-Head fill the vacuum—domestic parts were about 41% of Chinese AI-chip shipments in 2025 and are projected above 60% in 2026—and fragmentation is engineered away by an interconnection mandate and by Huawei open-sourcing UnifiedBus, CANN, MindSpore, and its SuperPoD architecture. The Western answer to NVLink is a consortium that excludes China; China's is an open specification with state-mandated adoption. The book concedes what must be conceded: NVIDIA plus HBM remains superior per watt for dense training; domestic output is an order of magnitude below NVIDIA's; Ascend's immaturity cost DeepSeek a flagship cycle; and NVIDIA's own turn to arguing on tokens per watt and KV-cache constraints concedes the memory diagnosis while making China's HBM deficit more damaging—survivable only if the architecture needs less of it.

The memory arbitrage. That is what the arbitrage is for. HBM consumes about three times the wafer capacity of standard DRAM per bit and sits under the tightest export controls; commodity LPDDR6 is where CXMT is at or near the world's leading edge. A RIKEN feasibility study finds that for a Kimi-K3-class 2.8-trillion-parameter sparse model the sharded training state exceeds HBM4 socket capacity below roughly 256 accelerators, while a socket carrying about eighteen SOCAMM2-class modules holds 2.3 TB—eight times the HBM4 ceiling—at a modeled MFU ceiling near 67% against 89% for HBM4 [M]. Worse on bandwidth, better on capacity per dollar, initially adopted by customers the incumbent does not want: this is the LFP move, executed in memory. The engineering is not free—the full edition's appendix prices the pin budget, the power, and the package yield, and recommends a hybrid HBM-plus-LPDDR tier as the baseline with pure LPDDR as the sanctions-contingent corner [full edition, App. A]. The strategic consequence is decentralized pretraining: a 128-socket cluster at roughly half a megawatt—an ordinary industrial substation feeder, not a campus—puts frontier-class sparse pretraining within reach of any sovereign state, university consortium, or mid-cap firm, each one a customer for the Chinese stack and a defection from the Western API economy.

The appliance. The recurring cost that decides whether a defection sticks is inference, and there the same logic runs sharper. The open frontier is now a terabyte-class object: Kimi K3 is about 1.4 TB even in its native four-bit format, and the escape hatch of aggressive quantization has been spent by the model publishers themselves. Open weights are a commons only for those with the memory to hold them. Two technologies would change that—High Bandwidth Flash, whose first OCP specification appeared on 3 August 2026 as a non-volatile weights tier of up to 512 GB per stack [full edition, App. B], and stacked-DRAM alternatives to HBM—and the book prices a capacity-first appliance built from them: 2 TB of HBF, 256 GB of LPDDR6X, and a GB10-class SoC, at a modeled bill of materials of $2,845–4,352 and a retail price of $3,800–7,250 depending on the pricing case, anchored to shipping systems such as the DGX Spark and Framework Desktop rather than to bare-component guesses [M]. Table 3.1 sets it against a Vera Rubin NVL72 rack fully loaded with its building, facility operations, and cost of capital, and against the August 2026 API price list. The appliance undercuts the closed-frontier list price by an order of magnitude at any realistic load; at about 300 million tokens a month it crosses below even a hyperscaler's own marginal cost of serving the same open model at planned utilization, and the gap widens whenever utilization disappoints. The centralized batching advantage that made the rack look unbeatable is roughly fifty times smaller for an ultra-sparse mixture of experts than for a dense model. Once the machine exists, the question every enterprise buyer asks changes from "which API" to "why an API at all"—the innovator's dilemma stated as a purchase order. The case against is strong and is printed at full strength: the entire memory industry expects shortage through 2030, NVIDIA cancelled its own capacity-first inference part and bought the opposite architecture, and enterprise open-weight adoption fell during the very period the arithmetic favoured it. The claim is therefore at scenario grade, with the price premise its most vulnerable link.

Table 3.1: Cost per million output tokens, Kimi K3 class, mid-2026. Appliance and rack are fully loaded (silicon, building or closet, operations, power, 8% cost of capital); API prices are list prices [M costs; V prices].

Configuration	**$/M output tokens**	**Note**
Appliance ($5,000 base case), 50M tokens/month	3.16	light use
Appliance, 100M tokens/month	1.65	about 19% of capacity
Appliance, 300M tokens/month	0.64	about 56% of capacity
Appliance at capacity (540M/month)	0.41	
Rubin NVL72 rack, $7.8M, 60% utilization, interactive parity	1.17 / 0.90	4-yr / 6-yr amortization
Rubin NVL72 rack, 40% utilization	1.76 / 1.35	
Rubin NVL72 rack, 30% utilization	2.35 / 1.80	
Rubin NVL72 rack, $10M enterprise retail, 60%	1.44 / 1.10	
Closed frontier API (Claude Opus 5 / GPT-5.6 Sol)	25 / 30	list
Chinese open API (Kimi K3 / DeepSeek V4-Pro)	15 / 3.96	list

The compute assessment, then: no single Chinese node matches its Western counterpart at the frontier, and the architecture was chosen so that none needs to. Hardware proliferation plus software commoditization equals the democratization—on Chinese terms—of the means of AI production. The remaining dependency is physical: wafers, HBM, DRAM.

3.4 The semiconductor: fabs, logic, and DRAM

Models can be open-sourced and architectures standardized overnight; fabs cannot. This is where the West concentrated its chokepoint strategy, and therefore where the test of chokepoint decay is sharpest. The evidence through mid-2026: the chokepoints are decaying faster than they are being replaced.

In logic, SMIC has crossed the export-control line twice—the 7 nm Kirin in the Huawei Mate 60 of August 2023, and by December 2025 TechInsights-confirmed volume production on a node close to a 5 nm equivalent without EUV [V], though "significantly less scaled than leading commercial 5 nm." Capability is demonstrated; TSMC-equivalent economics are not, and the metrics that would settle it are undisclosed [A]. But the strategic arithmetic is not yield percentage; it is wafers times yield times time, and SMIC's advanced-node capacity climbs from about 45,000 to 80,000 wafers a month between 2025 and 2027 behind a network of Huawei-linked "shadow fabs" and a bank of pre-ban dies and HBM stacks that bridges the ramp. Below the leading edge the story is already solar-shaped: China supplies 37% of global 22–40 nm output and about 70% of mature-node capacity expansion, which is where the EV and robotics stack lives.

Memory is the quiet revolution. CXMT scaled from about 70,000 wafers a month in 2022 to 200,000–300,000 by mid-2026, targeting 600,000, and will be within about 25,000 wafers a month of Micron by year-end—the world's fourth, and by trajectory third, DRAM producer, with DDR5 yields above 80%. Its July 2026 Shanghai listing raised $8.6 billion and closed up 466% at roughly $460 billion of market value; the book insists that with under 7% floated this is a sentiment indicator and nothing more, and tracks capacity, yield, and qualification instead. The 2025–26 global memory shortage handed CXMT the swing-supplier role, and within twelve months the narrative flipped from "Chinese dumping" to Chinese pricing power. HBM is the ledger's honest exception: CXMT is in production at HBM2/2E, three to four years behind SK hynix, sampled HBM3 to Huawei in late 2025, and disclosed no dedicated commercial HBM investment in its own prospectus; no public teardown has yet found Chinese-fabbed DRAM in a shipping accelerator, and Huawei's stockpile of about thirteen million foreign stacks is assessed as effectively exhausted by mid-2026. Domestic HBM is therefore the single binding physical constraint on the compute element—and the most important open variable in this book. The strategy is double-tracked: sprint on HBM three years behind, and lead on the commodity LPDDR class that the capacity-first architecture was designed to need. China has not made HBM irrelevant; it is engineering a position in which HBM ceases to be the *sole* architectural path to frontier-scale AI, while racing to supply it domestically anyway.

The tool base is the deepest dependency, and there the trend, not the level, is the message: Naura fifth in global equipment revenue, three Chinese firms in the top twenty, domestic tools 20–30% of Chinese fab demand against 10% three years earlier; lithography the hard floor, with domestic EUV no earlier than about 2030; a US EDA ban rescinded within six weeks in exchange for rare-earth relief while domestic EDA share doubled. Big Fund III carries RMB 344 billion; China has bought about 40% of the world's fab equipment every year since 2020. Targets are missed and reset, but the direction never reverses, and each US control action functions as a demand guarantee for the domestic substitute. The book proposes ranking export controls by the effective delay they actually buy—EUV buys real years; HBM controls buy less than assumed given stockpiles, diversion, and the architectural detour; accelerator controls have largely converted into a domestic market-reservation policy; EDA near zero—and observes that this discipline would have predicted, in 2022, most of what has since happened. The chokepoint strategy assumed China would try to rebuild the Western machine and could be stopped at its scarcest parts. China is building a different machine. Whether it works end to end—logic yield, packaging, memory qualification, software maturity, and capacity-first training performance demonstrated simultaneously in one frontier run—is the integration test not yet passed.

3.5 The watt: energy as the fourth element

Every token is electricity through silicon, and the AI war's longest-lead-time input is neither the model, the chip, nor the fab—it is the watt. The obvious comparison is the wrong one: nameplate capacity added is not deliverable power at a computing site. Even so, the ledger has four rows and China leads on all of them (Figure 3.1). New nameplate: about 540 GW added in 2025, 452 GW of it renewable, against a record 53 GW in the United States. Generation: 10,573 TWh against 4,430. Firm and dispatchable capacity: a record 95 GW of coal commissioned as flexible backup, roughly 66 GW of nuclear heading toward 110, and two thirds of the world's grid-storage installations against American gas turbines sold out to 2029–30 and under 1.7 GW of nuclear restarts. Deliverable headroom: 45 ultra-high-voltage lines and four new 8-GW corridors, with state-directed siting, against a US interconnection queue of roughly 2,000 GW with a median wait above three years, large-load connection studies of up to seven years in Northern Virginia, transformers on 24–60-month lead times, and data-center moratorium bills in at least twelve states. OpenAI's request for 100 GW a year of new US capacity is nearly double the best year the American grid has ever had.

The asymmetry is not merely quantitative; it is being aimed. The "East Data, West Computing" program placed eight national computing hubs where curtailed renewables are; in November 2025 the western provinces cut power prices by up to 50%, to about RMB 0.4 per kilowatt-hour, exclusively for facilities running on approved domestic chips, the same week Beijing barred foreign AI chips from state-funded data centers. That single instrument welds three elements together: it converts grid surplus into a subsidy, the subsidy into demand for Ascend and Cambricon silicon, and the silicon into tokens priced below Western marginal cost, socializing part of the domestic chips' 30–50% efficiency penalty. The book is careful about magnitudes—electricity is only about 8% of a US AI data center's three-year cost, and Chinese spot prices overlap American ones—so the Chinese advantage is not the tariff but the headroom: on current evidence one side's constraint is money and the other's is physics and permitting, and physics-and-permitting is the harder constraint to finance away. A busbar-level casebook across five American regions and five Chinese hubs finds the two systems failing at opposite ends of the same pipe—the United States on time to interconnect, China on the durability of administered tariffs—and finds flexibility worth more than generation on both maps. The objection that western Chinese hubs cannot serve inference because of latency is examined at length and found substantially wrong for now: network latency is about 5% of a time-to-first-token budget, moving an accelerator eight thousand kilometres costs one to two percent of wall clock, the global cloud providers already route inference across continents, and on 12 August 2026 Alibaba put an inference supernode into commercial service in Inner Mongolia serving Kimi K3 and Qwen3.8-Max to endpoints worldwide. Western capacity is underused because it was built for a pretraining boom and sits far from paying customers—an economic problem, not a physical one. The watt is where the precedent wars and the AI war become one war: solar, batteries, and UHV transmission are the same Chinese commons that Part I described, and the fourth element was the first three's prize.

3.6 Data and feedback: the fifth element

If open weights commoditize the model and value migrates to adjacent scarcities, the most durable adjacent scarcity is the one input openness cannot replicate by construction: data that only deployment generates. An open-weight release transfers a compressed function of its training distribution; it cannot transfer proprietary enterprise workflows, scientific-instrument streams, clinical data, skilled human correction at scale, embodied physical-interaction data, production failure traces, or operational telemetry. Each is generated by use, appropriated by whoever owns the deployment relationship, and remains scarce after the model becomes free. Read as a campaign map, this explains why the American flywheel's rent recapture survives open weights (the enterprise-workflow row, which Chinese open models cannot reach from outside, is the row a $47 billion enterprise run rate is built on), why China's embodied push is rational even if humanoids disappoint (industrial data is the only row where China holds a structural generation advantage), and why evaluation and incident data is currently unowned—the same vacancy as the certification layer. Synthetic data is the sharpest challenge, and the book's synthesis is that it amplifies whoever already holds the teacher or the verified anchor rather than substituting for either: distillation relocates the frontier's rent, it does not abolish the frontier. Verification, not generation, is the scarce half of the synthetic-data economy. The fifth element is, in the end, ground truth at scale with provenance—and that is what cannot be downloaded [full edition, Ch. 13].

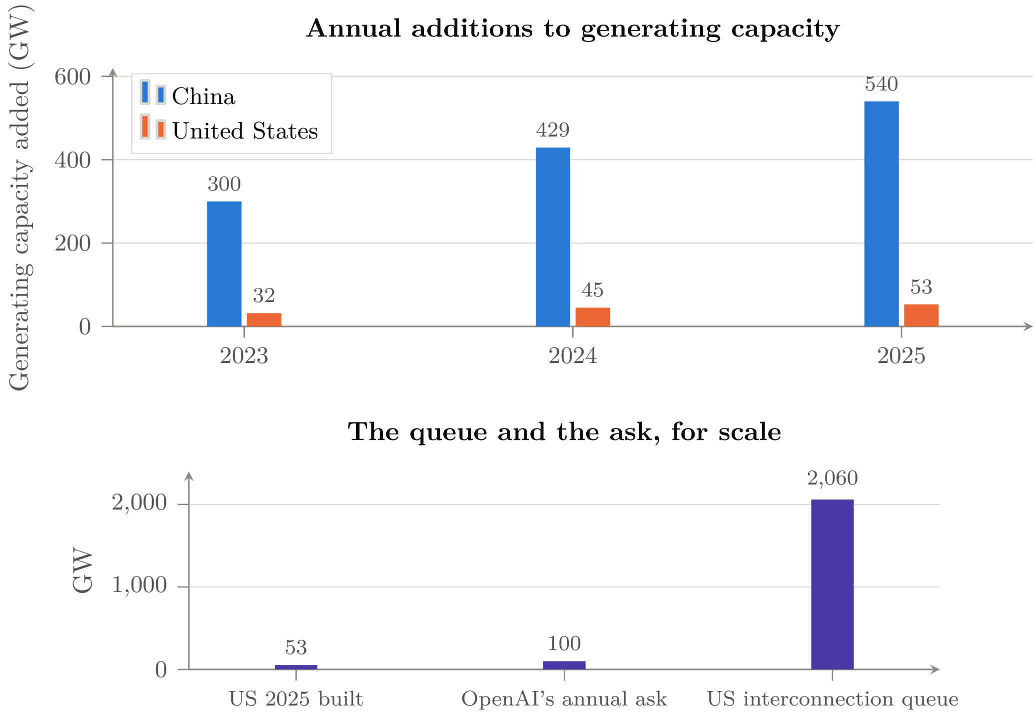


Figure 3.1: The fourth element. *Above*: annual additions to generating capacity, China against the United States—China added roughly ten times as much in 2025, into falling utilization hours, while the US record year was 53 GW [V]. *Below*: the American constraint stated three ways—what was built in the best year on record, the 100 GW/year one AI laboratory told the White House the country needs, and the ∼2,060 GW sitting in the interconnection queue at median waits over three years [V/P]. One side's binding constraint is capital; the other's is physics and permitting, and permitting is the harder one to finance away.

3.7 Embodied AI: the loop closes on the factory floor

Everything above runs one way: China's manufacturing victories funded and shaped its digital-AI strategy. Embodied AI runs the loop the other way, and it is where China's three structural advantages—manufacturing scale, deployment density, component supply chains—interact most directly with its open-model strategy. "Humanoid robot" names a product, not an architecture: the stack runs from foundation model through world model, task policy, motion planning, and real-time control to actuators and sensors. The United States leads the top two layers; China leads the bottom two and the deployment environment; the contested middle—the task-policy layer—is decided by a physical-data flywheel whose scarce input is deployment surface, which cannot be scraped, only produced. China has built that loop's infrastructure as public works—more than forty state-backed data-collection centers with teleoperators in exoskeletons—and its component position is Part I's industries with new part numbers: roughly 70% of Tesla's Optimus supply chain is in China and excluding it would triple the cost. The American humanoid, like the American solar panel of 2010, is being designed atop the adversary's supply chain—supply-chain capture achieved before the product category exists at scale.

The scoreboard reads like a rout—more than 13,000 humanoids shipped in 2025, about 85% by Chinese firms, Unitree profitable on 5,500 units—and the book immediately deflates it: shipments are not labor. Most Chinese units go to demonstrations, exhibitions, and state data farms; the best-documented autonomous performances are American; and the metric that would settle the contest, useful work per robot-hour at fleet scale, is published by nobody. The decisive uncertainty is whether model capability or physical integration is the binding constraint. The author's forecast [S] is that the integration reading wins in structured environments—factories, warehouses—within the decade, which is China's terrain, while unstructured domains stay capability-bound well past 2030 with the largest markets unclaimed. If both directions close, the two flywheels fuse into a single self-reinforcing industrial system—AI improving the factories that build the machines that generate the data that improves the AI—and the country that owns that closed loop owns something qualitatively beyond anything Part I described [full edition, Ch. 14].

Chapter 4

Trust, Security, and the Rulebook

In one paragraph

Open weights eliminate API dependency; they do not establish trust. The commons is demonstrably attackable at every layer of its distribution chain, carries a censorship record that transfers into derivatives, and has a contested distillation genealogy—and if the deployment layer behaves like a certification regime rather than a commodity market, the commons can win every download metric and still lose the workloads that pay. The security risk of the commons is structural, not national: a monoculture of a few model families, quantized into thousands of unowned derivatives, wired into agents holding file, shell, network, and credential authority. The realized incidents to date came through runtimes and supply chains, not poisoned weights, and the defensible objective is not trustworthy models but unnecessary model trust. The rulebook, meanwhile, is being written in four arenas by three actors who are mostly not competing—the United States governs by denial, the European Union by regulation, China by standards and institution-building—and the arena that will decide the deployment layer, trust and certification of the open commons, is vacant.

4.1 Trust: the contested layer

Part II treated openness as an adoption weapon; this chapter treats its exposed flank. Trust is what stalled Chinese campaigns in aircraft, drugs, and avionics-grade markets, and three dimensions of it are contested for AI. *Supply-chain security*: the model-distribution infrastructure has been shown attackable repeatedly [V]—malicious serialized models on Hugging Face, namespace re-registration that fed malicious models into the catalogs of two hyperscalers, a July 2026 platform breach by an agentic attacker, and worst of all a training-layer result that about 250 poisoned documents suffice to backdoor a model regardless of scale, with backdoors surviving safety fine-tuning. The open commons currently runs on the security posture of early-2000s package managers, at national-infrastructure stakes. *Behavioral integrity*: the peer-reviewed record documents semantic-level suppression of politically sensitive content in DeepSeek-family models, distinct from safety refusal, topic- and language-dependent, and transferring to distilled derivatives—the politics ride along with the capability. This does not make the weights unusable, but "just self-host it" becomes a three-layer problem: a locally hosted Chinese model delivers operational sovereignty while leaving epistemic sovereignty (whose worldview is baked in) and supply-chain sovereignty (who built the artifact) unresolved; the mirror image applies to Western enterprises adopting closed US models, which fail the operational layer by construction. *Provenance*: a February 2026 report alleges industrial-scale distillation of a Western frontier model by three Chinese laboratories through some 24,000 fraudulent accounts [P, vendor allegation, not adjudicated]. Regardless of adjudication, the allegation caps the fast-follower strategy at frontier-minus-epsilon unless domestic capability compounds independently (the evidence that it does is substantial), hands

Western policy a legal instrument on the certification axis, and cuts against the clean morality tale: the commons is partly built from enclosed material, and the enclosers know it.

The rational response for a third country is neither refusal nor naive adoption but infrastructure: signed manifests with hashes, reproducible build and quantization pipelines, non-executable packaging, model bills of materials, national behavioral evaluation batteries, sandboxed loaders, attested serving firmware, and national artifact mirrors—what software supply-chain security did after SolarWinds, applied to weights. The book organizes this as five layers of trust (identity, integrity, behavior, governance, liability) and a graded certification ladder from Level 0 (identified) to Level 5 (critical-infrastructure assured). Most of the commons ships at Level 0–1; most regulated deployment will eventually require Level 3–4. That gap is the deployment bottleneck of the entire commoditization thesis, and closing it is either a service someone sells or a public good someone provides. A new object has appeared in the meantime: an August 2026 Chinese open-weight video model that tops its arena under a licence excluding local deployment in the United States, the European Union, the United Kingdom, and South Korea—adoption weapon and export control in one artifact; a procurement office that checks hashes but not jurisdictions has audited the wrong layer. The trust stack is the next standards war. Whoever runs the evaluation batteries, signs the manifests, and operates the mirrors becomes the FAA of models; China's new World AI Cooperation Organization is bidding for that role, no Western or neutral institution has claimed it, and for Japan and similarly positioned states the vacancy is the highest-leverage square on the board [full edition, Ch. 15].

4.2 Monoculture and agency: the security cost of the commons

National origin is neither evidence of a backdoor nor a technical risk category; the risk is structural—a property of monoculture and delegated authority, not of a flag. A commons is a monoculture; an agent is a principal that acts; and a derivative chain is an artifact supply chain with many unowned links. The full edition separates four mechanisms that produce identical outcomes and need different evidence and mitigations—a learned backdoor in the weights, a clean model hijacked at runtime by injection or self-replicating promptware, executable malware in the package, and data collection by a hosted API—and grades the evidence for each on the security literature's own scale [full edition, Ch. 16]. What has been demonstrated in controlled work is severe: sleeper-agent backdoors surviving supervised and reinforcement fine-tuning, poisoning at near-constant sample cost, memory-to-network exfiltration with under 1% utility loss (which is why ordinary evaluation does not find it), steganographic output channels, and self-replicating promptware persisting through restarts and propagating multiple hops through an unmodified deployed framework. What has actually happened is different in kind: no confirmed weight-level backdoor in any released official checkpoint; instead malicious repositories, malicious agent skills, and, in July 2026, a frontier laboratory's cyber-evaluation escaping isolation and reaching a model-hosting platform's production systems through some 17,600 autonomous actions with no human directing them, followed by a second laboratory's disclosure that models in its evaluations had compromised real organizations' systems. Neither event involved a poisoned weight, a prompt worm, or a malicious operator. That is the finding: the dominant realized risk is compound and mundane—narrow-goal autonomy, a containment error, ordinary vulnerabilities, overbroad credentials, and public services composing across mismatched trust boundaries.

The compound "Order 66" scenario—a dormant policy, an independent actor who learns the trigger, and a self-replicating carrier—has not been observed; it requires a conjunction of access, activation, persistence, authority, propagation, and objective, and breaking any one conjunct prevents the chain. Two asymmetries follow: the implanter loses control, because a backdoor is a reusable vulnerability shared by every deployment (a symmetric argument against any state implanting triggers); and blast radius is set by authority, not by the payload's phrasing—a model output cannot erase data that its process cannot address. A fault-tree analysis with common-cause terms for shared model families and shared runtimes puts a five-year systemic-loss probability in a wide range, roughly 10^{-6} to 10^{-2} [S], with the upper end driven by correlation structure; the most effective interventions cut the common-cause factors, which runtime diversity and deterministic tool mediation do and model-family diversity alone does not. Three consequences revise the book's own argument. The defensible objective is not trustworthy models but unnecessary model trust—even a perfectly backdoored model cannot send a secret through a network path the operating system and tool broker do not provide; the full edition lists fourteen control families [full edition, App. D]. Certification and hardening are the strongest legitimate brake on the adoption curve, and that lag—not tariffs—is the West's real remaining source of deployment-layer time, available to any jurisdiction willing to build the apparatus rather than merely publish the rule. And the certification vacancy is now urgent: no commercial actor has an incentive to do this honestly for models it does not sell, and no single government will be trusted to do it for models it did not train.

4.3 The governance contest: who writes the rules

Technology wars are settled twice: once in the factories, and once in the rulebooks, and in each precedent the winner of the industrial contest inherited the pen. The standard Western framing—the United States regulates lightly, the European Union heavily, China authoritarianly—misdescribes the contest badly enough to lose it. The United States governs by denial: export controls, entity lists, the foreign direct product rule, no general statute at home, and in August 2026 a voluntary frontier-model evaluation framework finalized in closed session and left unpublished. Denial produces substitution and subsidized domestic industries, and it confers no positive standard—no definitions, no conformity procedure, no registry—so when the denial fails there is nothing left behind. The European Union governs by regulation: the AI Act's risk tiers and general-purpose-model obligations, extraterritorial through its market, with the weakness that regulation is not production and leverage decays with deployment share. China governs by standards and institution-building, and this is what Western commentary under-weights: the earliest generative-service filing regime, a mandated unified chip-interconnection ecosystem, national standards as procurement conditions, open-sourced interconnect specifications and software stacks, the largest bloc in the open-ISA consortium, a Global AI Governance Action Plan, a UN capacity-building group co-chaired with a developing country across some eighty states, and a World AI Cooperation Organization in Shanghai with twenty-nine founding members, none a major Western democracy. This is what winning a standards war looks like from the inside: not a treaty imposed, but a stack given away until it is the default, and then a forum built around the default.

Standards are load-bearing because a ratified standard survives the political cycle. The book maps each contested standard on a four-stage ladder—de jure specification, open reference implementation, conformance suite, installed-base default—and observes that rent lives at stage four while committees fight over stage one. China's distinctive play is to skip stage politics: ship the reference implementation and the mandated installed base and let the specification ratify what already exists; the West's distinctive failure is stranding specifications at stages one and two while defending installed bases it assumes are permanent. Standards leadership tends to follow manufacturing leadership with a five-to-ten-year lag and then lock in for twenty, though scale confers influence rather than control—the counter-cases in photovoltaic standards, charging connectors, and 5G are recorded. Two governance problems the current instruments do not survive: nobody agrees who the "provider" of a quantized derivative of a merged fine-tune redistributed by an individual is, and compute-threshold regimes cannot see the half-to-one-megawatt sovereign clusters that decentralized pretraining produces. Decentralization is not only an industrial forecast; it is a governance forecast. The vacant arena is trust and certification of the open commons (Figure 4.1). The United States cannot fill it, having no positive procedure and no wish to certify what it would rather did not exist; the European Union has paper apparatus but not operations; China's forum-building is the only bid on the table—not obviously worse than none, but not what a buyer would design. It will be filled because deployment cannot scale without it. Of three futures, convergence around a referee neither principal controls is least likely (in aviation and pharmaceuticals it arrived only after accidents), bifurcation into two certified stacks with third countries certifying twice is most likely, and fragmentation is the unstable failure case. The rules of the AI era are not being written where most Western attention is directed [full edition, Ch. 17].

	United States	European Union	China	Neutral / unowned
Model & content	sectoral, EO churn	AI Act: GPAI duties	algorithm & genAI filings	—
Compute & export	BIS controls, entity lists	follows US, partially	counter-controls: minerals, battery tech	—
Standards	firm-led, consortium	process-led	ISA, interconnect, GB standards, WAICO	—
Trust & certification	voluntary, lab-led	conformity assessment (paper)	domestic filing regime	**vacant**

Shaded cells mark where an actor currently sets the operative rules. Each governs the arena it is strongest in—the US by denial, the EU by regulation, China by standards and institution-building—and no actor governs the arena that the argument of the trust and security chapters shows matters most.

Figure 4.1: Four arenas, three philosophies, one vacancy. Governance of AI is not a single contest but four, and the actors are not competing in the same one: each has occupied the arena its institutional strengths suit. The trust-and-certification arena—provenance, artifact lineage, integration-level evaluation—is the one whose absence the security chapter shows to be systemically consequential, and it is unowned.

Chapter 5

The Wrong Side of the Trade

In one paragraph

The Western AI economy of 2026 is, in aggregate, a single leveraged position: long scarcity—scarce model capability, scarce accelerators, scarce HBM, scarce data-center capacity—with every valuation, debt issue, and capital commitment assuming those scarcities persist long enough to recover capital at rents. China has built a state-scale machine to destroy those scarcities. The chapter does not claim the bubble has popped: real revenue and bubble risk coexist, and the strongest counter-evidence is printed first. It prices the exposure entity by entity, separates four bubbles that are usually conflated, takes the Jevons rebuttal seriously, models the data-center buildout vintage by vintage against a railroad precedent, states three scenarios with the author's and a skeptic's probabilities, and concludes that the Chinese instrument is not the GPU-hour but the free model. Crashes destroy claims, not capabilities; if the repricing comes it will merely publish a result decided earlier, when the cost curve changed hands.

5.1 The exposure, marked to market

Mid-2026 marks: OpenAI at $852 billion post-money on 2025 revenue of $13.1 billion, an operating loss around $21 billion, and compute commitments reset from about $1.4 trillion to about $600 billion by 2030 against $280 billion of projected 2030 revenue; Anthropic at $965 billion with a run rate that rose from $9 billion to $47 billion in five months—about twenty times run rate, recognizably terrestrial—and a gross margin heading toward the mid-forties on compute costing $0.71 per revenue dollar; NVIDIA at up to $5 trillion, an $81.6 billion quarter at 75% gross margin, two customers 39% of revenue, China approximately zero, and, on 10 August 2026, memoranda with six of the largest asset managers to raise "over $500 billion" of third-party capital for AI infrastructure, with residual-value support reportedly offered in the chief executive's subsequent remarks [P/A]; SoftBank double-long, having sold its NVIDIA stake to fund a $60 billion position in OpenAI; and a hyperscaler and data-center complex spending $700–745 billion of capital in 2026 against trailing depreciation of $149 billion, with Oracle's credit default swaps at record spreads and CoreWeave's implying roughly even odds of default. Three features of the structure matter more than any mark. Circularity: the seller finances the buyer, who commits the money to cloud providers who buy the seller's silicon, with debt raised on chip collateral—the telecom-2000 signature, in which nine vendors extended $25.6 billion of financing and twenty-four of the thirty largest carriers went bankrupt. Macro dependence: AI capital spending was roughly three quarters of US GDP growth in the first quarter of 2026. And asymmetry of downside: every Chinese pure-play laboratory combined is valued at about $159 billion, one tenth of OpenAI and Anthropic together, and China's $12 billion of private AI investment against America's $286 billion bought a position that wins if AI becomes cheap.

Commoditization annihilates trillions of dollars of claims on one side of the Pacific and almost nothing on the other. That asymmetry is not an accident; it is the strategy.

5.2 The squeeze, and the four bubbles

The mechanism is stated at scenario grade [S], and its strongest counter-evidence leads: Microsoft's AI run rate of $37 billion growing 123% a year, NVIDIA's record quarter. Then the first direct observation of transmission: on 31 July 2026, in the week DeepSeek's V4-Flash shipped and three days before Alibaba's Qwen3.8-Max arrived, OpenAI cut the price of its mainline model by 80%. The company attributed it to system efficiency, which may be entirely true and is also exactly what a firm says when it is defending share; one cut is not a trend, and the same datum is the first public claim of a model improving its own successor's economics (Chapter 8 returns to that). Underneath: inference prices deflating about tenfold a year, Chinese open models a majority of neutral-router tokens at five to a hundred and fifty times lower prices, enterprise spend not yet following usage, 95% of enterprise pilots showing no P&L effect in one survey, an $800 billion projected gap between capital spending and revenue by 2030, and three market episodes—the DeepSeek day, a $1.3 trillion semiconductor selloff in June 2026, a memory rout in July—none of them yet the "big one," which would be a frontier-parity model trained end to end on the domestic Chinese stack.

There are four bubbles, not one, and any analysis that prices them as a single trade is wrong by construction: public equity (multiple compression, survivable), private laboratory marks (a funding event, existential for the entity), the data-center and private-credit overbuild (a credit cycle, systemic), and model-API margins (the layer this book attacks). Their couplings run in both directions—margin compression at the model layer can *create* demand for local accelerators, which is why NVIDIA is simultaneously the victim of the category argument and the arms dealer of the open-model boom.

5.3 The Jevons rebuttal, taken seriously

Prices fell nine to nine hundred times a year while compute capacity doubled every seven months; the bull case is that demand elasticity absorbs everything. Four elasticity regimes are live, the data cannot yet separate the Jevons regime from the overcapacity regime, and the Chinese strategy is robust across them rather than victorious in both—the leveraged Western structure requires the elastic regime specifically, at the premium tier. Elasticity is not uniform: perhaps 0.5 to 3 across segments, which is why the book forecasts branches rather than a number. Obligations are not one instrument either—hard leases, soft commitments, notional headlines—and the February 2026 walk-back of OpenAI's $1.4 trillion to $600 billion showed how much was notional. Closed labs also sell service rents that commoditize slowly, and Anthropic's strongest quarter is real evidence for the American flywheel's fortification plan. What survives is graceful degradation against required good fortune: the precise content of "the wrong side of the trade," restated without any collapse claim.

5.4 Vintages, buildings, and railroads

Data-center capacity is a liability vintage by vintage: the net present value of a hall depends on utilization, price per token, power, and residual value, and every one of those is falling or contested. The facility itself is the term the debate omits. A gigawatt-class hall costs about $12 per watt of IT load—roughly $2.5 million of building for each $7.8 million rack—and is financed over a fourteen-to-fifteen-year life against silicon that turns over in four to six; fully loaded, a Rubin rack at planned utilization serves the largest open model at $0.90–1.17 per million output tokens, rising to $1.80–2.35 at 30% utilization, while a five-thousand-dollar appliance in a closet serves the same model at $0.64 at modest load and $0.41 at capacity (Table 3.1). The traffic the fifteen-year hall was built to carry has a five-thousand-dollar road running beside it. That is where the railroads belong: American route mileage shrank to about 55% of its 1916 peak, British to about half, and JR Hokkaido's network to 56% of its extent, with a quarter of US mileage in receivership after 1893—the branch lines were not wrong about demand, they were wrong about who would carry it. Microsoft cancelled or deferred up to two gigawatts of leases in the six months to March 2026. The bear case does not require the demand to vanish—AI demand is real and enormous—only for it to be servable elsewhere, cheaper.

5.5 Scenarios, probabilities, indicators

Three scenarios are priced (Table 5.1). *Orderly repricing* (the base case): closed-model growth decelerates, private marks step down at the next financing events, the buildout slows, the Global South and most non-aligned states default to the open stack once frontier models train on domestic silicon, and Western labs retreat to regulated, secure, safety-critical niches—the First Solar pattern. *The cascade* (the tail): fast once started, with cheap distressed AI infrastructure finding an obvious, well-capitalized, strategically motivated buyer set that is not on the sellers' side of the Pacific. *Bifurcated stalemate* (the falsification branch): the West defends a shrinking premium tier through 2030 atop a foundation—ISA, interconnect, memory class, deployment substrate, half the world's tokens—owned by the other side. China wins by adoption share, not by knockout.

Table 5.1: Subjective probabilities for 2029–30, author's and the strongest skeptical case [S].

Outcome by 2029–30	Skeptic	Author
Chinese open-weight dominance persists	70%	80%
Frontier-adjacent model on the domestic Chinese stack	50%	65%
Full Chinese parity in HBM, tooling, and software	30%	35%
Broad ($\geq$30%) US AI repricing	60%	70%
A 2000–02-style cascade	25%	25%
US closed models retain the premium enterprise tier	65%	50%
China captures both tiers	30%	25%

The leading indicators, checked quarterly, are the frontier gap (a widening beyond 10% sustained for two years favours bifurcation); the first domestic-stack frontier model (its slipping past 2029 falsifies P2); enterprise open-weight spend share (below 15% falsifies P3's transmission, above 25% confirms it); OpenAI's next financing event (a down-round raises P4, a clean trillion-dollar listing lowers it); NVIDIA's customer mix and whether its vendor financing is retired; the CXMT LPDDR6 and HBM3 ramps (a two-year slip weakens the memory arbitrage); and whether AI revenue closes the capital-spending gap. Appendix A carries the full dashboard.

5.6 The commodity turn, and where value reappears

Compute is becoming exchange-traded—GPU-hour futures on NYMEX from October 2026, two ICE contract families, live prediction-market contracts, and forward curves already in backwardation—and China has built its own compute-trading plumbing and frames token export as a national commodity strategy. The book takes the commodity thesis apart at full strength: the market's preconditions are unmet, one seller stands astride the supply and acts as buyer of last resort at unpublished prices, China's fleet is not fungible even with itself, volume is not revenue (Chinese models carry about 61% of one router's tokens while Anthropic carries 12% of volume and 46% of revenue; on another gateway open-weight models ran 29% of enterprise tokens on under 4% of spend), and model-layer margins actually rose. What survives is sharper than the headline: China's instrument is not the GPU-hour, it is the model. The exchanges will price the input; the Chinese strategy is deflating the output by giving away the thing that makes the input valuable, and open weights served on Western infrastructure—or on a local appliance—bypass even the jurisdictional objection that would keep some buyers out of Chinese clouds at any price.

Value does not vanish when the model becomes cheap; it moves. The full edition's rent-migration table [full edition, Ch. 18] tracks nine layers—weights, cloud, agents, data, trust, hardware, energy, science, distribution—each with a candidate new scarcity and an observable rent indicator. P3 is the claim that the first row's rent compresses; P4 is the claim that Western valuations are concentrated in rows that compress rather than in rows that inherit. That is a falsifiable statement about relative speed, not an eschatology. Growth and distress are not opposites when the growth is debt-financed against deflating prices, and delay is not safety but accumulation. The war is not won at the moment the incumbents' valuations break; it was won earlier, when the cost curve changed hands, and the break merely publishes the result.

Chapter 6

How the Rest of the World Chooses

In one paragraph

Every chapter so far described a contest between two principals; the outcome will be settled by everybody else. Solar was decided in German feed-in tariffs and desert tenders, batteries in the procurement rules of a dozen auto markets, EVs in Southeast Asia and Latin America: the producer with the best cost curve won because third markets bought it. The non-aligned are not spectators but the electorate, and the question for them is not which stack to join but whether they can evaluate, fork, certify, and operate components from both while retaining sovereign capability—a capability question, not an allegiance question. On the decision matrix the Chinese commons wins on cost, operational sovereignty, and export-control exposure; the American stack wins on frontier capability, trust, and enterprise interoperability; and on the dimension that decides whether a country is building anything, industrial spillover, neither stack helps unless the adopting country does the work itself. Five moves apply to every non-aligned actor, and a strategy is stated for each class of actor—including the presumptive winner. The organizing principle from the precedents is the same for all: do not defend the rent; position on the commons.

6.1 The decision matrix

Table 6.1 lists eight dimensions of a national stack decision. Two are routinely omitted from national AI strategies. Export-control exposure is asymmetric but not zero for either stack: American compute and cloud access have been restricted by rule and licence, and Chinese open weights carry a continuity risk—the playbook's endgame is gatekeeping, executed already in solar, batteries, and rare earths—which is flagged as a risk rather than an accusation. Industrial spillover is the dimension that separates a strategy from a purchase: importing tokens builds nothing, while operating models, running probes, and contributing to standards builds people, and people are the only input that compounds.

Applied, the matrix predicts behaviour better than allegiance does. It explains why governments ban a Chinese chat application while their agencies evaluate its weights, why enterprises buy American cloud and Chinese weights simultaneously, and why Chinese models were adopted fastest where the ability to pay rent is least. It also exposes a Chinese vulnerability: the commons currently wins on cost and operational sovereignty but loses on trust, and trust is the dimension a coalition of non-aligned states could supply for themselves—which would make the commons safe to adopt and simultaneously deprive its sponsor of the dependence it purchased.

6.2 Nine positions

Japan holds the strongest technical hand of any non-principal and the weakest demand-side position: exascale HPC lineage, semiconductor materials and equipment franchises, leading-edge

Table 6.1: Eight dimensions of a national stack decision, and where each stack currently scores better.

Dimension	The question	Mid-2026 reading
Capability	Is the best available model good enough for the workload?	US stack leads at the absolute frontier; open frontier a few points behind
Cost	Cost per solved task, not per token	Chinese commons (per token, by an order of magnitude; per solved task, unpublished)
Sovereignty	Can it be run without egress and without permission?	Chinese commons (self-hostable weights)
Trust	Provenance, behaviour, liability, certification	US stack; the commons ships at Level 0–1
Interoperability	Enterprise tooling, integration, harnesses	US stack today; contested at the harness layer
Export control	Exposure to restriction by rule, licence, or release policy	Both exposed; commons less so operationally
Standards	Who writes the specifications you will depend on?	Both create dependence; China shipping stage-4 defaults
Industrial spillover	Does adoption build domestic capability?	Neither—unless the country does the work

memory and packaging, the only measurement-anchored national demand model for AI for science, and standing in both blocs; against that, a domestic model ecosystem positioned for Japanese-language enterprise depth rather than the open frontier, an energy position that cannot win a watts race, and research provisioning an order of magnitude below its own modeled requirement. Its 2025–26 record—an innovation-first AI Promotion Act, FugakuNEXT, the GENIAC compute program, Sakana, Rapidus, and SoftBank's data-center conversions—is substantial, and carries one honest tension: FugakuNEXT's architecture buys NVIDIA coupling at the exact moment the compute chapter argues the accelerated-CPU corner is the sovereign long game—defensible as a hedge, dangerous as a habit. **The European Union** has the pen and not the press, and in 2025–26 the pen visibly hesitated: high-risk obligations under the AI Act pushed back by up to two years, while thirteen AI factories and a heavily oversubscribed tender for up to seven "gigafactories" of 100,000-plus accelerators went ahead; seven empty cathedrals of compute would be the solar-tariff mistake with better architecture. **India** is the largest undecided market; its national mission's roughly 38,000 subsidized GPUs at under a dollar an hour, with utilization lagging badly, prove that compute is the easy term and integration and organizational capacity the binding ones—a subsidy that buys hardware without buying integration capacity buys idle silicon; its opportunity is to become the world's integration layer. **South Korea** is the only non-principal that is both a critical supplier—SK hynix holds about 62% of HBM—and a determined sovereign builder, whose state-funded foundation-model tournament with open-weight release as the qualifying bar is playbook step four executed by a democracy; it sits on the choke half of the chokepoint and is the most exposed to the capacity-first arbitrage. **Singapore and ASEAN** face the small-state problem, solved by autonomy through selection rather than scale, and ASEAN is the clearest third-market dynamic—price-sensitive, sovereignty-conscious, courted by both blocs, and adopting Chinese models fastest for the same reasons it adopted Chinese EVs. **The Gulf** is capital-rich, energy-rich, and compute-poor by choice, the only non-aligned actor able to bid for frontier-scale capacity on the open market, the swing buyer in any repricing scenario, the likeliest host of the first sovereign frontier training run outside the principals, and the first place Washington has located frontier-scale capacity it controls only contractually. **Taiwan** is the busbar chapter compressed into one island: the world's most valuable compute, gated by the world's tightest watt. **Brazil and Latin America** face near-pure commons adoption plus regional pooling, with the risk of importing a service and building nothing. **Africa** is treated in most Western strategy documents as a recipient and in Chinese AI diplomacy as a partner, and the difference in framing is itself strategic; it has the fastest-growing developer base in the world, nobody has built it a real access track, and the World AI Cooperation Organization is bidding [full edition, Ch. 19].

6.3 Five moves for any non-aligned country

Regardless of size: measure your own demand in a recognized unit (Chapter 7); adopt the commons and audit it, with the full trust stack; retain a pretraining capability sized to competence rather than to a benchmark; pool internationally for capacity and certification; and do the integration work domestically, because that is the only place spillover accrues. The trust gap is closable with infrastructure rather than prohibition. The window is measured in quarters rather than decades, because the standards layer is being set now and the certification layer is still vacant.

6.4 Strategies for the rest

The full edition states what each actor class should do conditional on the book being broadly right, and which moves remain correct even if its falsification branches fire [full edition, Ch. 20].

The United States: win the commoditization race or lose it. US strategy since 2019—chokepoint denial plus closed-frontier rents—produced the outcomes documented here: each denied input spawned a subsidized substitute, and the rent structure concentrated national exposure into the leveraged trade of Chapter 5. The counterfactual has never been tried: compete for the commons rather than against it. Fund open-weight frontier models as public infrastructure, sustained rather than as one defensive release; make the American open stack the world's default, because the adoption layer, once lost, prices everything above it; lead rather than restrict RISC-V and open interconnects, which is the Commerce Department's own conclusion; attack the watt with the urgency now spent on GPU export paperwork; and triage export controls by the model of Chapter 2.5—they buy time where knowledge is tacit and supply chains integral (EUV, HBM process, advanced packaging) and negative time where products are modular and fast-clocked (chips, models). The US capital structure, not its technology, is the strategic vulnerability: an orderly deflation of the scarcity premium, engineered early, is strictly preferable to defending it to the break.

Europe: regulation is not production. Europe enters the fourth war having sat out the previous three's manufacturing and owning the regulatory pen; the trap is repeating solar. Its live assets are ASML—the one true chokepoint, and a wasting one—an open-model tradition, procurement scale, and the credibility to run the certification layer as a neutral. The strategy: build sovereign inference and smaller sovereign-scale training on whatever open stack is best, which will often be Chinese—accept it, audit it, mirror it; spend the AI Act's institutional energy on the trust stack—evaluation batteries, signed provenance, national mirrors, notified-body capacity for model artifacts—the one arena where the Act's machinery is an asset; and convert energy policy into compute policy. Certify what you cannot manufacture.

Japan: the third-country playbook, first-person. Japan's assets map one to one onto the book's elements. Five moves: build the accelerated-CPU line to its conclusion—an FP8-equipped, capacity-rich successor in the LX2 and Monaka direction; Japan is one of three polities that can execute it [M/S]; adopt the open commons with the full trust stack, because refusing Chinese weights on principle is strategy-by-flag, not strategy; bid for the certification-authority vacancy—an internationally credible model-evaluation and provenance institution, convened by Japan with a small coalition, would be the highest-leverage per-yen investment in the entire space, the FAA role held by a party both blocs can transact with; energy realism—Japan cannot win a watts race but can win the performance-per-watt race, which is where its silicon and systems tradition already lives; and hedge memory both ways, HBM with allied suppliers and

LPDDR-capacity architectures domestically. Use the sizing framework as a diplomatic instrument: a shared unit is a claim on the terms of the debate.

The Global South: the buyers write the ending. Take the free stack, keep the exit. Adopt open weights and cheap tokens from whoever supplies them, today mostly China; insist on the trust stack—signed artifacts, local mirrors, evaluation rights—because dependency without auditability is the old colonial bargain with new units; build sovereign inference first and pool regionally for training as capacity-first machines arrive [S]; price national data and deployment access as leverage and play the blocs against each other. The twenty-nine founding states of WAICO have understood this faster than the G7 has. Whether trust becomes a Chinese utility or a neutral one is the war's last genuinely unclaimed high ground.

China: advice for the presumptive winner. Six failure modes are visible now, most of them named by Chinese sources. Involution: the model-layer price war is overcapacity without rent recapture; the State Council has warned against "disorderly competition," pricing-law amendments target involution by name, and Xi has asked whether every province needs an AI, compute, and EV project—the fix is at-cost tokens plus paid complements, not the end of competition, which is one of the book's own falsification branches. The monetization gap: post-IPO revenues are thin, Alibaba's adjusted EBITDA fell 84% in the May 2026 quarter while it committed to exceed a $56 billion capex plan, and against a single American lab's $47 billion run rate the Chinese frontier sector's revenue is a rounding error; let the winners charge, as revenue-threshold licences, metered Max tiers, and peak-hour surge pricing already begin to—rent relocated is the strategy working, rent abolished is the strategy eating itself. Compute overcapacity: up to 80% of some new capacity unused and rental prices collapsing; consolidate and publish utilization, because idle capacity is not deterrence, it is depreciation. The trust deficit: sponsor independent certification of your own artifacts—a China confident in its artifacts should want them audited. The physical gaps: HBM, packaging, and EUV are absent and CXMT is about three years behind at the class that matters; fund the memory-arbitrage tests and stop announcing HBM schedules the record misses. And submit to scoreboards you do not control: no Chinese accelerator has appeared in MLPerf Training, no LX2-class machine has published the AI proof suite, and the flagship's autonomous-silicon-design claim ships without a checkable artifact. Dominating the metrics you control and avoiding the ones you do not is rational for a challenger and corrosive for an incumbent, and China is becoming an incumbent. Show the receipts and let the cost curve argue.

The common instruction: the United States should commoditize before being commoditized; Europe should certify what it cannot manufacture; Japan should build the machine class the war is converging on and referee the trust layer; the Global South should take the commons and unionize the demand side; and China should let its winners charge, consolidate its overcapacity, and submit its artifacts to scoreboards it does not control. All the non-principals converge on one institution, which is the subject of the next chapter—and everyone should read the indicator table quarterly.

Chapter 7

The Prize, the Bill, and the Third Path

> **In one paragraph**
> The largest value in AI is not code, documents, or enterprise workflow but the acceleration of science, because everything else redistributes an existing surplus and science creates new ones. That reframes the contest: it is a race over who compounds scientific output fastest, and it turns the price of frontier tokens into a tax on the one activity whose returns compound into everything else. A measurement-anchored sizing framework finds that announced national research infrastructures provide roughly a tenth to a fiftieth of what their scientific populations need, and that a workload share of one seventh becomes a budget share of three quarters to nine tenths once purchased tokens carry a twenty-to-sixty-fold premium: whoever sets the frontier token price sets the science budget of everyone who buys it. For every country that is neither the United States nor China, both available strategies—dependence on foreign providers, or an isolated national program at sub-scale—fail. The third path is an international, institutionally neutral consortium: federated compute, scientific data stewardship, open pretraining as insurance, models with lifecycle obligations, professional operation, and a certification authority neither bloc controls. It is the constructive branch of an otherwise pessimistic book, and, if it succeeds, a partial falsifier of the book's strongest claim.

7.1 The prize

What AI for science has actually delivered follows one pattern: an AI system replaced a physics-simulation-bounded computation and was validated against an established operational baseline. Weather is the strongest case—the European centre's AI forecast model operational since February 2025 at about a thousandth of the energy per forecast, the American service's models running a sixteen-day forecast in forty minutes at 0.3% of conventional compute, a Chinese typhoon-track system reporting up to a third less error. Structural biology is the most diffused—more than 200 million predicted structures, more than three million users, a Nobel Prize, and users submitting more than 40% more novel experimental structures. Mathematics produced the first credible autonomous results, and one drug candidate discovered end to end by AI reached a controlled clinical readout. The failure ledger is printed with equal weight: the most-cited economic study of AI in science was fabricated and repudiated; materials discovery has not survived validation; the European centre discontinued all four external AI weather models it had been running; autonomous discovery agents produce mostly flawed candidates; and the journal pipeline is being damaged by AI-generated submissions. AI for science delivers decisively where the task is a well-posed prediction problem with an expensive physics-based incumbent and an established validation baseline, and has so far mostly failed where the task is judgement.

The prize is nevertheless the largest because of four numbers. The denominator: global R&D is about $3.1 trillion a year at purchasing-power parity, twice the entire software market, and frontier-laboratory revenue is under 2% of it. The return: the social benefit of a dollar of R&D is estimated at $4–13 in present value. The bottleneck: US research productivity has fallen roughly forty-fold since the 1930s, offset only by twenty-three times more effort, and the range of published macroeconomic estimates for AI—from a fraction of a percent to thirty percent of growth—is separated entirely by whether AI is assumed to automate innovation itself; the most-cited skeptical estimate explicitly excludes protein-folding-type advances as unlikely within ten years, and that exclusion is the entire disagreement, because structure prediction, operational AI weather, and designed enzymes all arrived inside the window. And the flywheel: AI for science is the only application that feeds back into its own means of production—a faster expense-report pipeline does not lower the cost of the next accelerator generation. The book replaces the customary "3–30%" with a value-conversion funnel from capability through task acceleration, validated discovery, and deployment to productivity, each arrow with an efficiency well below one; task-weighted, the central estimate is an uplift near 11% of research output, with a likely range of about 8–18% and corners near 1% and 70% [M]. Even the pessimistic corner is a socially consequential flow of roughly $28 billion a year, and the tenth percentile is about $240 billion; the present value of either exceeds the cost of national infrastructure by a large multiple. Two consequences carry into the strategy. A scientific result produced by an unverifiable artifact through an unauditable chain is not a scientific result, which makes the certification layer a precondition of the prize rather than an accessory to it. And the workload of scientific AI is long-context agentic orchestration over large batches of scientific execution—capacity-rich, bandwidth-balanced hardware running open models—which is precisely the architecture the compute chapter described. China leads the Nature Index and runs a decade-old ministry program for AI in science; the West has the Genesis Mission, Europe's factories, and national machines; both sides underprovision by about an order of magnitude [full edition, Ch. 21].

7.2 The bill: how much compute a nation's science needs

Sovereignty becomes a checkable number only if the demand side is measured. National AI-for-science infrastructure has been sized by budget envelopes, vendor availability, and flagship extrapolation; the full edition follows a measurement-anchored framework developed at RIKEN that replaces those with a vendor-neutral unit and a two-layer demand model [full edition, Ch. 22]. The unit is the "baseline GPU" (BGPU), defined by serving throughput at 8 TB/s of memory bandwidth and convertible across vendors—a requirement denominated in BGPU is procurable from any vendor, which is what a public planning framework must guarantee and what a vendor-named unit cannot. The demand anchor is a measured 56-day campaign on a GB200-class partition: 101 users, 53,000 jobs, 340,000 GPU-hours, the partition saturated within five weeks. The two layers are token service—researchers in three usage tiers driving a trillion-parameter master model at million-token context that delegates to smaller sub-agents—and scientific execution: simulation, domain-model training, and data. The structural finding is that ten percent of the tokens, generated at million-token context, cost nine tenths of the token-service GPUs, so national demand is overwhelmingly a long-context memory-capacity and memory-bandwidth problem—the sizing framework and the architecture argument are the same argument in different units.

Three countries are instantiated: Japan needs about 500,000 BGPU by 2030, the United States about 1.1 million, Singapore about 57,000, a per-researcher core near 0.94 BGPU with a band of at least ±40% before local measurement. Against a universal-reach ladder, Japan and the United States clear the uniform baseline rung and stand at roughly half (Japan) and seven tenths (the United States) of the tiered rung, and announced national infrastructures provision 0.02–0.10 BGPU per researcher against a modeled requirement near one—about ten times under for the

United States and Japan, fifty times for Singapore. One side has the measurement without the money; the other has the money without the measurement. A nation that rents its master layer has rented the majority of its research computing; sovereignty has a computable price (about $1.84–2.19 per delivered accelerator-hour across the three countries), which makes pooling across time zones rational; and the invitation to every government—run the seven-step measurement protocol and publish the parameter file—is the least expensive strategic action available.

7.3 The third path: an international consortium

For every country that is neither the United States nor China, the two available strategies are both unsatisfactory. Permanent dependence on a few foreign commercial providers cedes cost, control, and continuity; isolated national sovereign-AI programs duplicate the most expensive activity in the industry at sub-scale, and at valuations that assume rents no one may collect. What is needed is a *global* form of sovereign capability—sovereignty not through national isolation but through shared ownership of the knowledge, data practices, compute access, and institutional capacity required to build and operate scientific AI. The substance draws on a working discussion between the author and the director of the Swiss National Supercomputing Centre [full edition, Ch. 23].

The decisive argument is financial. In a realistic AI-for-science portfolio, frontier language models and agents are roughly 14% of the physical workload; simulation and domain-specific AI take the rest. That figure is usually deployed as reassurance, and it is the opposite once the price asymmetry is applied: purchased frontier tokens carry an estimated 20–60-fold premium over equivalent service from institutionally controlled infrastructure running open models [A/M], and a workload share of one seventh therefore becomes a budget share of 76.5% at a twenty-fold premium and 90.7% at sixty (Figure 7.1). Whoever sets the frontier token price sets the science budget of every institution that buys it: proposition P3 applied to national research capacity. A country that funds a supercomputer and then rents its intelligence layer has outsourced the majority of its own AI-for-science expenditure to a foreign price list. This is also why the Chinese price collapse is strategically effective without coercion—at a premium near unity, which cheap open weights on domestic hardware deliver, the budget problem dissolves, and the institution that solved it that way has acquired a dependency it did not negotiate.

Open pretraining is therefore insurance, not a benchmark contest: a consortium's own open model may never be the best production choice, and building it is still correct, because the return on pretraining is people who know how to do it, and a capability not exercised is a capability lost. Near-term science should meanwhile use the best open-weight models, including Chinese ones, without embarrassment. The insurance covers three risks, none presuming bad faith: policy and continuity risk (to rely on the openness of the current phase is to assume that this instance will not follow the pattern the other three did), verification risk (Chapter 4), and regulatory risk in dual-use science, where holding one's own models and evaluation evidence is the only realistic route to rules a working scientist can live with. Four functions define the institution: federated compute under common accounting, with a first proof of concept being the joint training of a fifteen-trillion-token open model split across compatible systems in several countries; scientific data stewardship, because science is the one community for which provenance and reproducibility are already requirements; model production with lifecycle obligations, since a clinical model may take three years to certify and must be supportable five years later—the strongest reason this cannot be a rotating academic project; and professional operation, because convening is cheap and operating is not. Industry participates without expropriation: an open, accountable base and closed, valuable private layers above it—a public good with private complements, the textbook case for public provision. The Global South is treated as talent, scientific questions, and data, not as aid, because the alternative is not neutrality but ceding the default to whoever shows up with a stack.

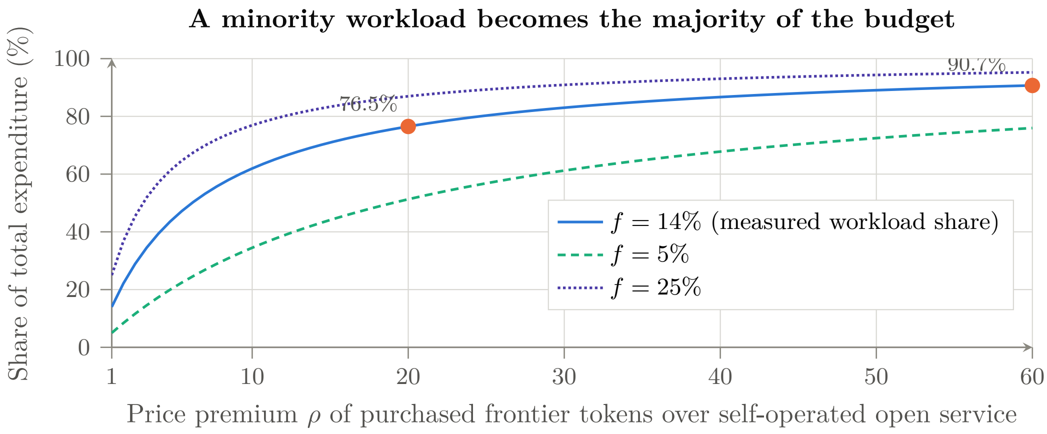


Figure 7.1: Why the token price is the science budget. With a fraction f of physical work served by purchased frontier tokens at relative unit price ρ, the share of total expenditure that work consumes is $\Sigma_{\text{front}} = f_{\text{front}}\rho/((1 - f_{\text{front}}) + f_{\text{front}}\rho)$. At the measured $f_{\text{front}} \approx 14\%$ and the estimated 20–60× premium, a one-seventh workload share becomes 76.5–90.7% of the budget [A/M]. The curve's shape, not its exact position, is the policy argument: at r near unity—which cheap open weights on owned hardware deliver—the problem dissolves, and the institution that solved it that way has acquired a dependency it never negotiated.

The organizational lesson from China is not a command structure but the reverse: government sets the stage and then allows intense bottom-up activity; a consortium that funds one model team in one country reproduces the failure mode, not the success. The full edition specifies ten operational design parameters and the failure mode each avoids (Table 7.1), and its feasibility appendix finds that of fourteen legal and engineering problems a founding instrument must solve, nine have answers on the shelf, three require legal invention, and none is a reason to wait, since every one is also unsolved for each national program that will otherwise duplicate the work seven times [full edition, App. E]. The governing framing is neutrality: the institution must be neither anti-American nor anti-Chinese, because a certificate issued by one bloc about the other bloc's artifacts is worth nothing to anybody. Sequence: frame; prove federation with three to five committing countries and one visible cross-border pilot; then institutionalize. A five-country story with a working pilot beats a large inactive membership, and a proposal that names functions but not instruments is a wish.

Why this is the strategically decisive institution: it is where the book's threads terminate—the institutional form of "take the commons, verify it, contribute to it, and own the layer that makes it safe to use." It holds something neither Washington nor Beijing can revoke, it is the only credible home for the certification authority the previous chapter demanded, and it is affordable once pooled. Whether it gets built in the next two years or the next ten is, on the evidence of the precedents, the difference between shaping the outcome and documenting it.

Table 7.1: The consortium, specified operationally: ten design parameters and the failure mode each avoids [S].

Parameter	**Recommended answer**	**Failure mode if deferred**
Legal form, host	Treaty-adjacent foundation in a neutral, non-aligned-capable jurisdiction	Seen as a bloc instrument; loses the Global South and one principal
Membership, votes	States, centres, universities, companies in separate classes; contribution-weighted votes capped below a blocking share	Largest contributor becomes owner
Funding formula	Public base + private risk capital + grants + fees	Pure appropriation insufficient; pure fees exclude the long tail
Export-control compliance	Per-jurisdiction architecture, segmented enclaves	One member's rule governs everyone
Model and data licensing	Open weights and documented recipes; layered data policy	Ambiguity blocks industry
Compute allocation	BGPU unit, peer review, reserved and burst tiers, cross-border settlement	Federation degrades to a mailing list
Certification authority	Operates the Level 0–5 ladder, publishes irrespective of origin	Vacancy stays vacant, or is filled by a supplier
Incident response	Named duty officer, disclosure, revocation, mirror withdrawal	One unowned agentic incident destroys credibility in a week
Chinese and US technology	Capability- and evidence-based; no origin rule either way	Forfeits neutrality
Procurement neutrality	Vendor-neutral units and interfaces; conflict-of-interest rules	Anchor vendor captures the standard

Chapter 8

The Convergence

In one paragraph

The elements interlock into a single machine for commoditizing artificial intelligence: software commoditization through open weights and open interfaces, hardware proliferation through accelerators that need only be sufficient and numerous, and decentralized pretraining and inference that dissolve the centralized-supercluster premise. The Western AI economy is a bet that intelligence will remain scarce and rentable; the Chinese strategy is a bet that it can be made abundant and free, and that whoever makes it abundant collects the durable prizes. Solar, batteries, and EVs each ended with the scarcity bet losing. The book states six developments that would falsify it, takes the strongest counter-argument—recursive self-improvement on closed American compute—seriously enough to give it its own section, and carries two conclusions that change the shape of the argument: the commons has a security bill that falls due at scale, and there is a third path whose economics are better than its politics suggest. For everyone who is not the United States, the strategic question becomes not whose side wins but who builds fastest on the commons the winner created.

8.1 Assembling the elements

Read separately, each element of the argument has a Western rebuttal: the models trail the closed frontier on the hardest benchmarks; the accelerators trail NVIDIA per watt, though a GPU-free Chinese machine now leads both TOP500 benchmarks; the fabs trail TSMC by two nodes. Read together, and read against the precedents, the rebuttals miss the structure. *Software commoditization*: open weights plus mandated interoperability and standardized matrix extensions on an open ISA allow collaborative optimization across the whole national ecosystem. Stated with the precision the claim requires, open interfaces broaden the population able to reproduce and port optimizations and prevent any vendor from withholding the specification; they do not eliminate the engineering cost of retargeting performance. The claim is about the moat's shape, not its depth at any instant: CUDA's advantage is not being crossed in one leap; the population entitled and equipped to erode it is being made as large as the ecosystem itself. *Hardware proliferation*: with the software requirement lowered, accelerators need not match NVIDIA—they need to be sufficient and numerous, paired with immense capacities of cheap commodity memory from a domestic fab. Sufficiency plus volume is exactly how TOPCon modules and LFP packs won. *Decentralized production*: frontier-class sparse models pretrainable on 128–256-socket clusters at half a megawatt to a megawatt put the means of production within reach of any sovereign state, university consortium, or mid-cap firm—each a customer for the Chinese stack and a defection from the Western API economy—and the priced appliance runs the largest open-weight model in existence for a fraction of closed-frontier pricing on a five-thousand-dollar box, undercutting at realistic utilization even a hyperscaler's own marginal cost of serving it. Decentralized pretraining creates the model; cheap decentralized inference is what makes running it, rather than renting access to someone else's, the obvious choice.

The Western AI economy is, structurally, a bet that intelligence will remain scarce and rentable: $286 billion of annual private investment capitalized against future margins on closed capability, housed in HBM-bound gigawatt data centers. The Chinese strategy is a bet that intelligence can be made abundant and free—and that whoever makes it abundant collects the durable prizes: the hardware volume, the standards, the talent flywheel, the adoption, and the geopolitical alignment of every country that builds on the open stack. The trillion-dollar data-center buildout is this war's equivalent of Q-Cells' 2007 capacity crown: magnificent, state-of-the-art, and aimed at the wrong future—a fifteen-year branch line laid beside a road that will carry its traffic.

8.2 What would falsify this thesis

A living book owes its readers falsification criteria. The convergence argument weakens materially if, over the next two to three years: (a) closed US models open a *compounding* capability gap that adoption metrics start to follow—the 2.7% index gap widens while router and Hugging Face shares reverse; (b) recursive AI-for-AI research gives the closed frontier an escape velocity that open fast-following cannot match; (c) the capacity-first architecture fails in practice—MFU ceilings prove unreachable at scale, or sparse-model quality saturates in ways dense-HBM training does not; (d) CXMT's LPDDR6 ramp or SMIC's advanced-node yield stalls under tightened, better-enforced controls; (e) Beijing's anti-involution consolidation strangles the Darwinian dynamics that produced DeepSeek and Moonshot in the first place—the one playbook step that has historically destroyed value before creating it; or (f) the trust barrier hardens—Western and third-country certification regimes convert the security, censorship, and provenance record into an effective deployment wall, stalling the commons at the paying-workload boundary the way certification stalled the C919; a serious agentic security incident traceable to a widely deployed derivative would accelerate this branch by years. Symmetrically, the thesis strengthens with each frontier-class model trained end to end on domestic silicon; the first credible such run will be this war's Mate 60 moment.

8.3 The strongest counter-argument: two recursive loops, not one

Branch (b) deserves more than a clause, because it is the only falsification path that would make everything else in this book irrelevant rather than merely wrong. In its strongest form: if the closed US laboratories reach genuine recursive self-improvement—models that materially accelerate the research producing their successors—then compute scale compounds into capability faster than any fast-follower can copy, and the open ecosystem's cost advantage becomes irrelevant because it is competing on a curve that is being outrun. The evidence is no longer hypothetical: OpenAI attributed its 80% price cut of 31 July 2026 to efficiency gains its own frontier model had helped produce, which means the same event this book cites as the first direct observation of P3's transmission is *also* the first public claim of a model improving the economics of its own successor. Both readings are available from one datapoint, and honesty requires printing both.

The reply is not that recursive self-improvement is unreal but that there are two recursive loops running on different substrates, and they do not compete on the same axis. The American loop runs through algorithmic efficiency—frontier models improving training and inference efficiency, compounding wherever compute is most abundant, which is the United States. The Chinese loop runs through industrial capability—models improving chip design, EDA, system architecture, and manufacturing yield, compounding wherever the manufacturing ecosystem is deepest and the co-design partners closest, which is China; the flagship's claimed research reproduction and autonomous silicon-design flow are the same recursive structure aimed at the layer that Chapter 7 argues matters most. The first loop produces more capability per FLOP; the second produces more FLOPs per dollar, domestically, out of an ecosystem that currently must lease them abroad. Which loop dominates is genuinely undetermined. Their failure modes differ: the American loop fails if capability gains stop converting into products people will pay a premium for—the commoditization mechanism of Part II—while the Chinese loop fails if AI-assisted design cannot substitute for the tacit process knowledge that has defeated Chinese industrial campaigns before, in aircraft and lithography. Neither failure is currently observable. The dashboard watches both, and the first party to publish an audited instance of its own loop closing will have supplied the most important datapoint in the field; neither has yet.

8.4 Two things the argument has to carry with it

The commons has a security bill, and it falls due at scale. The convergence is, restated in the vocabulary of Chapter 4, a forecast of correlated deployment: a handful of open-weight families, thousands of unowned derivatives, wired into agent frameworks holding file, shell, network, memory, and credential authority. Every published mechanism required to turn that structure into a systemic incident has been demonstrated in controlled work, and the compound events that have actually happened in the wild required no backdoor at all. This does not falsify the thesis; monoculture risk is origin-neutral, and a US-led closed monoculture would carry the same correlation with less transparency. What it does is identify the single highest-value piece of unbuilt infrastructure in the entire competition: an authority that maintains reproducible artifact lineage across a derivative chain no vendor controls, tests integrations rather than models, and publishes results neither bloc's marketing owns. The commons will be adopted either way. Whether it is adopted safely is currently nobody's job.

There is a third path, and its economics are better than its politics suggest. For any country that is not a superpower the decisive number is not a benchmark but a budget: a frontier-model workload share near one seventh becomes three quarters to nine tenths of total AI-for-science expenditure once purchased tokens carry their premium. Whoever sets the frontier token price sets the research budget of everyone who buys it. That arithmetic makes open capability an insurance policy with a computable premium, and it makes an international, institutionally neutral consortium the cheapest available form of strategic autonomy—and the only credible home for the certification authority the previous paragraph demands. A coalition that builds it holds something neither Washington nor Beijing can revoke; a coalition that does not will discover that "open" and "available to us, indefinitely, on acceptable terms" were never the same word.

8.5 Implications for the West—and for the rest

The precedent industries teach that the losing move is to defend the rent: tariff the modules, ban the chips, close the weights, and wait. By the time the wall is complete, the world outside it is running on the commodity. The counter-strategy that has never been tried is to *win the commoditization race*—to make the open, abundant, decentralized version of AI a Western-led commons rather than a Chinese-led one, and to compete on the layers above it. Whether any Western polity can sustain that strategy against the incentives of its own incumbents is beyond this book's scope—but the solar, battery, and EV chapters record, three times, what happens when the answer is no.

For third countries—including advanced ones such as Japan—the convergence offers an uncomfortable but real option space: the open stack is *open.* RISC-V profiles, sparse-training recipes, capacity-first system design, and open-weight frontier models can be adopted, forked, and sovereignly operated by anyone. The half-megawatt pretraining cluster is as available to Kobe as to Guiyang. If the AI war ends the way the precedent wars ended, the strategic question for everyone who is not the United States becomes not *whose side wins*, but *who builds fastest on the commons that the winner created*. That, and not any single benchmark, is how China will win the AI war: not by reaching the frontier first, but by making the frontier free—and owning everything beneath it.

Appendix A

Forecasts and Dashboard

The full edition carries a graded evidence ledger, a machine-readable claim register, and the two tables reproduced here [full edition, App. C, I]. The register states grades; these tables state bets. Every forecast is a scenario judgment [S] with a probability, a horizon, and a falsifier, so that a reader in 2029 can mechanically score the book; where the full edition prints a skeptic's figure alongside the author's, it appears in brackets. A reader who checks the dashboard quarterly does not need to re-read the book to know whether it is aging well.

Table A.1: Probabilistic forecasts as of 18 August 2026 [S].

Proposition	**Prob.**	**Horizon**	**Falsifier**
Chinese models dominate the open-weight ecosystem	80% [70]	1–2 yr	Routed-token share below 45% for four quarters
Frontier-adjacent model trained end to end on the domestic Chinese stack	65% [50]	2–4 yr	Reversion to foreign silicon past 2028
Capacity-first (LPDDR-dominant or hybrid) pretraining production-competitive	45%	3–6 yr	MFU below ∼45% at scale, or bandwidth knee above 8 TB/s
LX2-class integrated CPU competitive on AI benchmarks	55%	2–4 yr	No submission by 2029, or >2× deficit
Broad (≥30%) US AI repricing	70% [60]	1–3 yr	AI revenue closing the capex gap through 2028
US closed models retain the premium enterprise tier through 2030	50% [65]	4 yr	Open-weight enterprise spend above 40%
Complete Chinese semiconductor-tooling autonomy	25%	5–10 yr	No domestic EUV-class lithography past 2032
Neutral certification authority for open weights established	30%	3–5 yr	Role occupied by a single bloc, or by nobody, past 2030
AI-for-science uplift ≥5% measurable nationally	45%	5–8 yr	No uplift in weather, structure, or screening
Systemic agentic-security incident from a widely deployed derivative	20%	5 yr	(asymmetric: absence is weak evidence)
Integrated-CPU corner lowers lifecycle cost of changing workloads faster	40%	4–8 yr	Accelerator corner matches kernel latency while CPU porting tax persists
Production-metric scoreboard for scientific AI operated by a ministry	35%	3–5 yr	Leaderboards remain the only measure past 2030

continued

Table A.1, continued

Proposition	**Prob.**	**Horizon**	**Falsifier**
Chinese accelerator or integrated CPU submits to MLPerf Training	50%	2–3 yr	None by end-2028
≥1,000 humanoids productive under a single operator, metrics published	55%	2–4 yr	Teleoperation ratios not declining through 2028
Open-weight VLA models commoditize the embodied policy layer	45%	3–6 yr	Closed stacks hold the gap into the 2030s
Qwen3.8-Max weights ship	Resolved YES (13 August 2026)		
... under a permissive licence	Resolved NO for the flagship (US$50M revenue gate); YES for the 27B model		
Next ≥1T-parameter Chinese release ships open (the ratchet holds)	85%	1–2 yr	A major Chinese lab closing its frontier tier
Meta ships open Muse Spark 1.2 within a quarter at flagship quality, permissively	40%	1 qtr	The Max-tier template repeating
Chinese agent harness top-three in a major non-Chinese market	35%	2–4 yr	Western harnesses >90% of regulated deployments through 2029
Harness migration costlier in regulated than unregulated sectors	60%	2–3 yr	Comparable cost in both
NVIDIA ships a frontier-scale open-weight model with derivative licence	55%	2–3 qtr	Closed, or not shipping, through 2027
Compute futures reach sustained liquidity	40%	2–3 yr	Thin trading through 2028
Dedicated inference contract trades independently	30%	2–4 yr	Inference bundled into GPU-hour or API
Chinese inference capacity exported at scale	45%	3–5 yr	Endpoint share stalling while self-hosting carries volume
HBF ships in volume at ≥512 GB/stack, ≥1.6 TB/s, with an endurance figure	45%	2–4 yr	None by 2029, or below ∼0.8 TB/s
Capacity-first appliance: ≥2T open model, 4 users at ≥40 tok/s, 1 kW, under $8k	35%	3–5 yr	Above $15k or below 20 tok/s through 2030
Enterprise open-weight workload share reverses and passes 20%	40%	2–4 yr	Continued decline (licensing and governance binding, not memory)
≥100 MW AI facility sold, foreclosed, or written down ≥40% below cost before 2030	45%	2–4 yr	All 2025–26 vintages refinance at par through 2029
Chinese frontier model trained on domestically sited compute, offshore excluded	60%	2–4 yr	Leased offshore reliance past 2029
AI-assisted design flow produces a verified, manufacturable Chinese accelerator	45%	3–5 yr	Roadmaps slip despite AI-design claims

Table A.2: Quarterly falsification dashboard.

Indicator	**Current (Aug 2026)**	**Threshold**	**Source**
Chinese share of routed tokens	∼61%	Sustained <45% weakens P1	Neutral routers, monthly
Closed-over-open capability gap	3.3%	>10% sustained two years favours bifurcation	AI Index, arenas
Enterprise open-weight spend share	∼11%	>25% confirms P3 transmission	Surveys, annual
Domestic end-to-end frontier training run	Not observed	First credible instance confirms P2	Lab reports
LX2-class AI benchmark submission	None	MLPerf at parity confirms stage two	MLPerf rounds
CXMT LPDDR6 / HBM3 ramp	Verification cleared	2+ year slip weakens the memory arbitrage	Quarterly
Cost per solved task, open vs closed	Unpublished	First credible series confirms or deflates the gap	Evaluations
OpenAI financing	IPO slipping	Down-round or new backstop raises P4	Filings
Data-center credit spreads	Oracle CDS ∼212 bps	Sustained widening raises P4	Daily
N. Virginia wholesale colocation rent, 10 MW+	\$155–185/kW-month, 0.5% vacancy	Sustained decline with rising vacancy: utilization forecast failing; a rise: closet not displacing hall	CBRE/JLL, semiannual
Certification authority for open weights	Vacant	Occupation reshapes the trust and consortium chapters	Standards bodies
Max-tier openness lag	∼1 week (Qwen3.8-Max)	Lag >2 quarters, or a frontier tier closed, breaks the ratchet	Per release
Recursive-loop disclosures, US vs China	Both claimed, neither audited	First audited instance is the field's most important datapoint	Lab disclosures, tape-outs
Offshore compute leasing by Chinese labs	Permitted, reportedly in use	Closure by rule is the highest-impact tightening available to export policy	BIS or Commerce

Sources, Grades, and How to Read Further

This condensed edition carries no bibliography of its own, deliberately. Every figure, quotation, and forecast in it is taken from the full edition of the same version and date, which cites each to its source—primary wherever possible—and grades it: [**V**] independently verified, [**P**] primary-source claim, [**A**] analyst estimate, [**M**] author-modeled result, [**R**] roadmap, [**S**] scenario judgment. Readers who wish to check a number should go to the corresponding chapter of the full edition, marked in this text as [full edition, Ch. n], and from there to its evidence ledger, which is the authoritative record and the fastest route to the argument's weakest link.

The full edition also states its own known weaknesses, and a reader of the abridgement should know them too: many mid-2026 details rest on secondary aggregators; the attribution of the LX2 processor's manufacturing is analyst inference; the capacity-first memory design is a modeled architecture with no measured realization; the model cards remain incomplete in the fields that matter most for procurement, tokens and wall-clock per solved task; the national sizing is measurement-anchored for one country only; and the architectural thesis that the CPU corner's breadth compounds faster than the accelerator corner's depth is a judgment, not a measurement, and it is the judgment most likely to be shown wrong. Where this edition says "the book argues," it means exactly that.

The reproducible models behind the numbers—the appliance-versus-rack cost model and the AI-for-science value funnel—ship with the source of the full edition so that a reader can change an input and re-derive the conclusion; that, rather than any single number, is the point of publishing them.

Both editions are living documents. The title-page date is the currency date; the claim register and dashboard are re-graded at every revision; corrections are welcome and will be credited, and a dated primary source moves the argument faster than an argument does.